Beyond Averages and Snapshots

Connectivity Profiles and Daily Weather

Team Plot Twist • DAS732 Data Visualization, Term 1 (2026-27)

Programming Assignment 2 • gridMET: July-September 2023

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***Abstract*—What do averages and snapshots hide about inequality, relationships and change over time? We examine this question through separate country-connectivity and daily-weather investigations. Required Infovis experiments compare network layouts, interactive bundled parallel coordinates, hierarchy remodelings, horizons and ridgelines. Required SciVis experiments compare scalar mapping, contour intervals and quiver encodings, with three eight-frame GIFs. Selected extensions add matched-country change analysis, adoption quantile changes, exploratory NodeTrix groups, complete-month differences, weather intersections and persistence. In 175 matched countries, broadband/adoption rank association is 0.831 for levels but 0.054 for changes. In the sampled 92-day weather footprint, hot/dry overlap occupies 16.27 percent of area-days while a local run reaches 67 days. Adding a declared wind threshold narrows overlap to 2.57 percent. Twenty-nine figures document encodings, measured evidence, limitations and sensitivity. Results remain descriptive and restricted to their stated cohorts, spatial support and temporal aggregation.**

***Index Terms*—information visualization, scientific visualization, gridMET, parallel coordinates, treemap, NodeTrix, set intersections, temporal comparison**

I. Introduction

The central question is: **What do averages and snapshots hide about inequality, relationships and change over time?**

Assignment 1 examined whether mobile leapfrogging closed the digital divide. Assignment 2 extends the country investigation with network, multivariable, hierarchical and temporal methods, and separately examines daily gridMET weather. The shared question concerns what different visual representations reveal. The country and weather tables are not joined, and no relationship between connectivity and weather is inferred.

Our answer has three parts. Similar adoption levels can accompany different infrastructure profiles; country-level associations need not describe within-country changes. A rising adoption distribution can coexist with persistent regional differences. In weather, spatial averages, two-date differences and single-condition summaries can conceal geographic variation, overlapping conditions and long local runs.

The report combines required method experiments with selected analytical extensions. Each figure states its question, how to read its encodings, the measured evidence, the interpretation and its limits. Required browser interactions and scalar animations are supplied separately because a static PDF cannot demonstrate them.

Sources. [2], [15].

II. Data, transformations and comparison design

Country source and units. The input is the Assignment 1 country-only table derived from the Our World in Data/ITU indicators, with regional labels from the A1 lookup. Internet adoption is the percentage of people using the internet. Landline, broadband and mobile measures are subscriptions per 100 people. Subscription intensities can exceed 100; they do not count unique subscribers or imply universal ownership. Eritrea, Myanmar and Palestine remain Unclassified where the supplied regional lookup is missing.

Country cohorts. The 2020 graph has 179 complete countries and 590 undirected similarity edges. Parallel coordinates, the scatterplot matrix and the 2015-2020 change comparison use 175 matched complete countries. The 2020 treemap contains 184 countries with both user counts and adoption shares, totaling 4,521,767,530 included internet users. Temporal views retain 159 countries with adoption values in every year from 2000 to 2020; the six classified regional rows contain 158 of these countries. Denominators differ because required variables differ, and are not interchangeable.

Graph construction. For each of the four indicators, subtract its 2020 complete-case country mean and divide by its population standard deviation. Compute Euclidean distances and connect each country to its five nearest neighbors; take the undirected union. Edge similarity is 1/(1+distance). This derived graph measures profile resemblance, not cables, traffic, influence or geographic proximity. Weighted greedy modularity supplies an exploratory partition for NodeTrix. Sensitivity checks change k to 3, 5 and 8 and compare five Louvain seeds at each k.

Weather source and support. The team-supplied interval is July 1-September 30, 2023. gridMET fields are daily maximum temperature tmmx, precipitation pr, wind speed vs

and clockwise-from-north wind FROM-direction th. NetCDF packing is decoded before source Kelvin temperatures are converted to Celsius. Existing derived caches already contain Celsius. The displayed extent is 125 W-66 W, 24 N-50 N. Reading every sixth native grid cell gives approximately 0.25-degree sampled spacing; this is subsampling, not block averaging or native resolution.

Two temporal supports. Required scalar experiments and GIFs use July 1, July 14, July 27, August 9, August 22, September 4, September 17 and September 30, uniformly 13 days apart. Full-period extensions use all 92 days: 31 in July, 31 in August and 30 in September. The common footprint contains 13,416 cells valid across all four fields and dates. Missing cells remain masked. Eight-date global extrema refer to the eight sampled fields, not all 92 days, the whole Earth or a climatology.

Aggregation and thresholds. Domain area summaries use cosine-latitude weights on the sampled regular grid. Hot means daily Tmax at least 35 C; dry means daily rainfall below 1 mm. The new three-set comparison uses daily wind speed at least 5 m/s as an analyst-selected windy threshold. Alternatives are exported. These definitions describe daily gridded summaries, not drought diagnoses, heatwave warnings, gusts or population exposure. The dataset resolves neither atmospheric height nor within-day coincidence.

Conventions. Monthly temperature differences are September mean daily Tmax minus July mean daily Tmax. Monthly rainfall is the sum of daily precipitation, not an average. Wind roses retain the source FROM-direction. Quiver arrows show flow TO-direction, using u=-s sin(theta) and v=-s cos(theta); a north wind therefore points south. Arrow display bearings account for longitude scaling and the plotted aspect. Directions must not be averaged as ordinary degree values.
Sources. [16], [3], [4], [12].

III. JavaScript implementation and interaction evidence

Where JavaScript is used. The browser implementations are in `web/`. `web/app.js` implements parallel coordinates using native JavaScript and SVG, including region-centroid edge bundling, brushing, axis reordering, year/region controls and fixed matched-country scales. `web/treemap.js` uses vendored D3 v7 for zoomable treemaps and two table-to-tree remodelings. `web/network.js` uses D3 for the interactive node-link view; its three layout positions are computed by NetworkX and stored in JSON. Native JavaScript in the PCP satisfies the JavaScript language requirement; D3 is a library used by the other pages.

How to run. From the `ass2` folder, run `python run_visualizations.py`, or double-click `START_VISUALIZATIONS.cmd`. Open `http://127.0.0.1:8765/web/home.html`. The launcher links the report, gallery, three browser pages and animation files. A local HTTP server is needed because the pages fetch JSON. D3 is bundled locally, so the visualization pages need no CDN.

Parallel-coordinate demonstration. Open `web/index.html`. Keep matched countries and fixed axes enabled. Set bundling to 0% and then 50%; exact axis intercepts stay fixed. Drag vertically on an axis to brush a range, inspect the selected count, then clear brushes. Click an axis title to move it one position right. Switch 2015/2020 and region to inspect the same measurement semantics. Curves between axes are rendering geometry, not observations or temporal transitions.

Treemap demonstration. Open `web/treemap.html`. Start at the root, click a region, inspect the breadcrumb and Back button, and hover a leaf for counts and adoption. Change from region-first to adoption-band-first hierarchy and repeat from the root. Both remodelings preserve the same 184 country leaves and total user count. The two report screenshots show different zoom subsets; their displayed totals are not an equal-population comparison.

Network demonstration. Open `web/network.html`. Switch Spring/Spectral/Circular, search for India or an ISO code, and select a country to inspect neighbors and its original profile. Change the similarity threshold, then reset; filtering hides stored edges rather than recomputing country measurements. Pan and zoom to inspect crowded neighborhoods.

Validation status. JavaScript syntax, local asset loading, data counts and implementation handlers were checked for this revision. Existing screenshots show implementation states. No supported live browser session was connected during the revision, so the above end-to-end interaction sequence remains a manual check before submission. Screenshots alone do not verify interactions. `web/README.md` gives the exact source-file map and demonstration checklist.

Submission evidence. A2 explicitly requires JavaScript implementations of interactive bundled parallel coordinates and zoomable treemaps. It also requires an interactive node-link comparison of at least three layouts. This report locates the corresponding code and documents the controlled comparisons. A narrated video is not required by specification 1(f); three eight-frame scalar GIFs remain required animation artifacts.

IV. Country structure: distinguish the graph from its drawing

We first ask whether the apparent country groups are supported by stored connections and measured profiles. The required three-layout experiment holds the graph fixed. NodeTrix adds a way to inspect groups; the companion profiles explain their measurements. The group partition is explicitly exploratory.

A. Three layouts of one derived 2020 country graph

Question (Fig. 1). How much of a country similarity graph's apparent structure comes from its layout?
How to read the graph. Each point is one country with complete 2020 values for four connectivity indicators. Color gives its supplied region; a link joins countries selected by the standardized five-nearest-neighbor rule. The three panels use identical nodes and edges, but spring, spectral and circular algorithms assign different drawing positions.

TABLE I
REQUIRED IMPLEMENTATION EVIDENCE AND ITS LOCATION.

Requirement	Implemented behavior	Source / report evidence
Network	Three layouts, selection, filtering, zoom/pan	`web/network.js`; Fig. 1 and supplied browser page
JavaScript PCP	Region bundling, axis reordering, brushing, matched years	`web/app.js`; Figs. 4–5
JavaScript treemap	Zoom, back navigation, two hierarchy remodelings	`web/treemap.js`; Figs. 8–9
Temporal methods	Two horizon and two ridgeline variations	Figs. 12–15
Scalar methods	Palettes/normalization, global/local extrema, contours	Figs. 16–18; three eight-frame GIFs
Vector methods	Sampling density, proportional/equal arrow lengths	Fig. 25; source FROM to arrow TO conversion

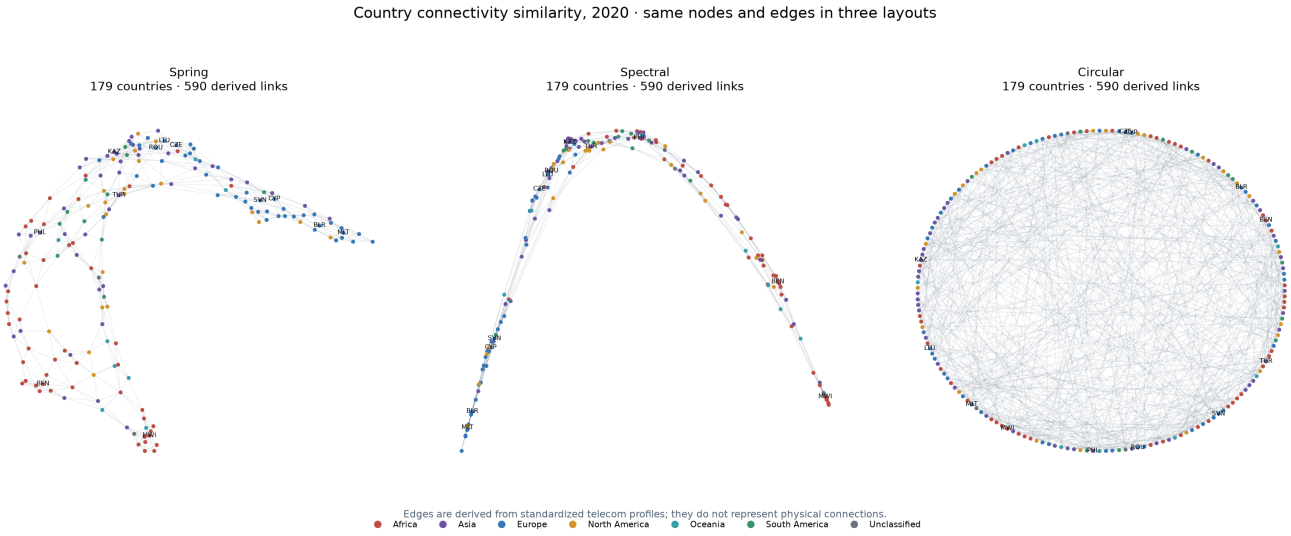


Fig. 1. Three layouts of one derived 2020 country graph.

Detailed interpretation. The key comparison is not which panel contains more groups: every panel contains exactly the same 590 edges. A neighborhood that looks separated in spring layout may be less visually isolated under spectral placement. Checking the actual links distinguishes graph structure from layout geometry. Slovenia's degree of 12 exceeds the mean 6.59, but this is a property of a symmetrized five-nearest-neighbor construction, not evidence of national influence or physical network capacity. This figure earns its place as the required controlled layout experiment.

Limits. Trace a country and its links across all three panels before calling an apparent group meaningful. A node's coordinates have no geographic or indicator units. Crossings in the circular view reflect placement; they are not additional connections. Regional colors permit comparison with supplied labels rather than proving discovered communities.

Takeaway. The same 179-country, 590-edge graph creates different visual neighborhoods under different layouts. Verify an apparent group using its links; geometric separation alone is not a discovered community.

B. NodeTrix overview

Question (Fig. 2). Which locally connected country groups and between-group connections are obscured by a node-link overview?

How to read the graph. Each colored square is a symmetric binary adjacency matrix of one exploratory group. Each row/column is a country in the same ISO-code order; a colored cell is an internal edge. Gray segments retain the 65 individual external edges and attach to their actual country row or column. Cell size is constant, and labels give group size and internal edge density. Colors identify derived groups, not regions.

Detailed interpretation. The weighted greedy-modularity partition gives seven groups of 40, 37, 35, 32, 14, 13 and 8 countries. There are 525 internal edges and 65 between groups, compared with 3.70% overall edge density. Group internal densities range from 15.6% to 64.3%. This fits a locally denser, globally sparse display, but high internal connectivity is partly expected from the nearest-neighbor construction and modularity objective. Changing k to 3/5/8 gives 13/7/5 greedy groups; alternative Louvain seeds also change memberships.

Limits. Use the figure to inspect this partition, not to claim seven natural country types. Matrix area grows with group size, so compare the density labels rather than colored area. Paths and links remain derived profile relationships. This is a static NodeTrix-style overview; the full interaction design of the original paper is not claimed.

Takeaway. Local matrix structure is inspectable, but the number and membership of groups depend on graph and optimizer choices.

C. Group profiles

Question (Fig. 3). Can nearly equal internet-adoption means conceal different fixed and mobile infrastructure profiles?

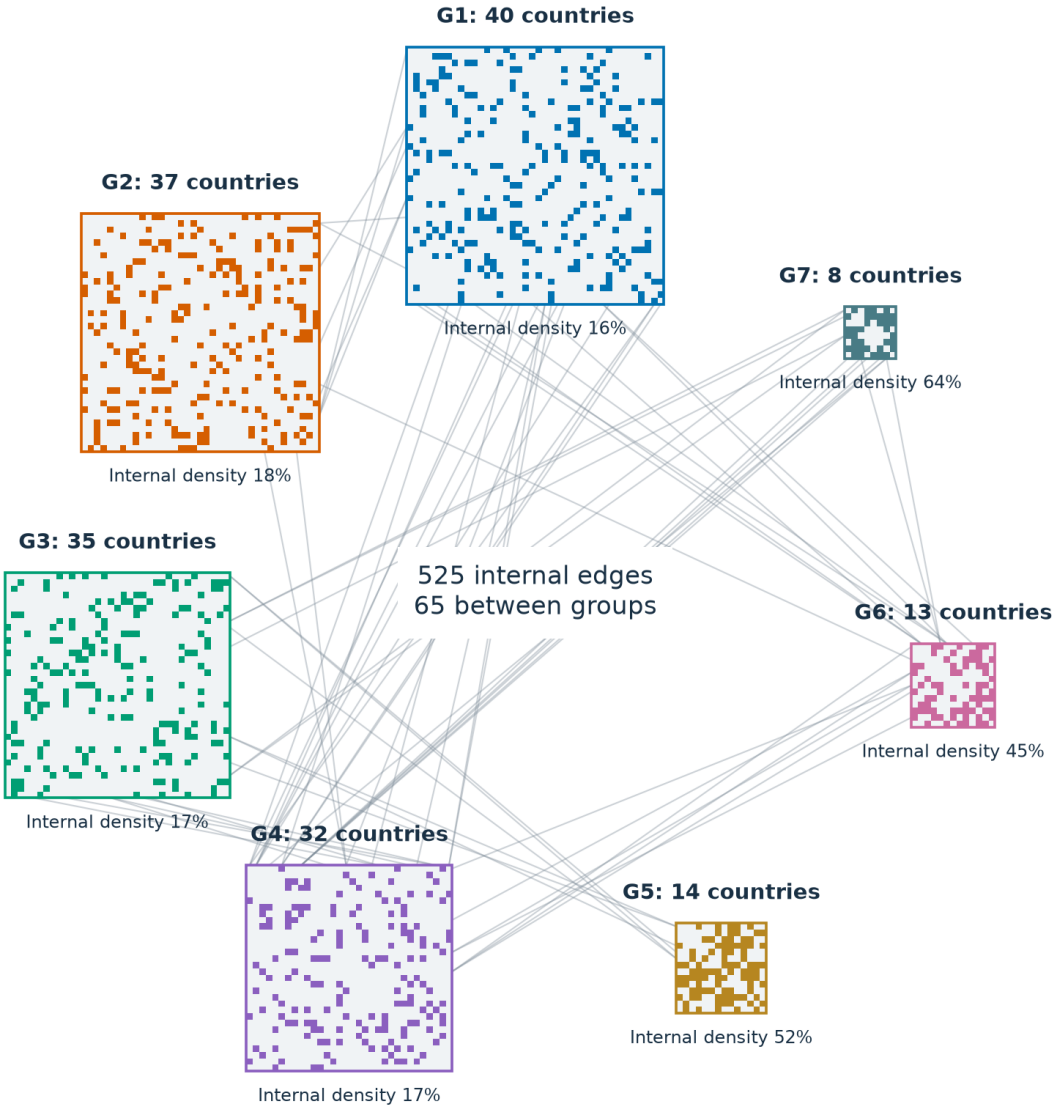


Fig. 2. NodeTrix overview. Method reference: [7], [8].

How to read the graph. Rows are the seven NodeTrix groups. Each panel uses its own raw-unit horizontal axis. A dot is the equal-country mean, while the thick translucent segment spans the 25th-75th country percentiles. These intervals show within-group heterogeneity, not confidence limits. Group colors and IDs agree with the network overview.

Detailed interpretation. G1 and G2 average 86.1% and 86.8% adoption, yet landline intensity averages 15.3 versus 41.2 per 100, broadband 24.6 versus 37.0, and mobile 140.3 versus 119.7. The adoption mean alone therefore conceals infrastructure differences. Groups were constructed from these indicators, so those contrasts describe the chosen partition rather than independently validate a causal technology pathway. Other rows preserve the broad profile range and their overlapping country spreads.

Limits. Group membership changes under k and optimizer sensitivity. Means and interquartile ranges are country-weighted. Mobile subscriptions are not unique users; 140.3 subscriptions per 100 does not imply 140.3% of people own phones. Panel scales differ and should be read separately.

Takeaway. Similar mean adoption accompanies different infrastructure profiles in this exploratory partition; a single adoption average cannot describe the complete profile.

V. Country Relationships: Inspect Levels, Changes and Hierarchies

The browser methods preserve individual measurements while supporting filtering and hierarchy exploration. The scatterplot matrix verifies pairwise relationships with position encodings. We then compare country-level associations with within-country changes and distinguish internet-user counts from adoption percentages.

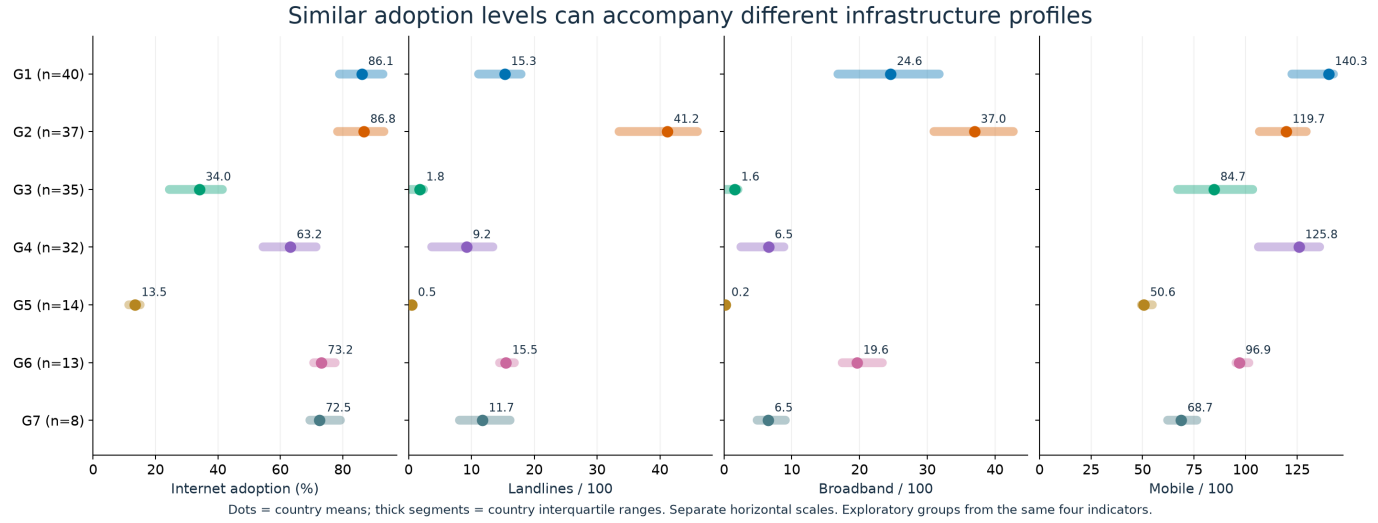


Fig. 3. Group profiles.

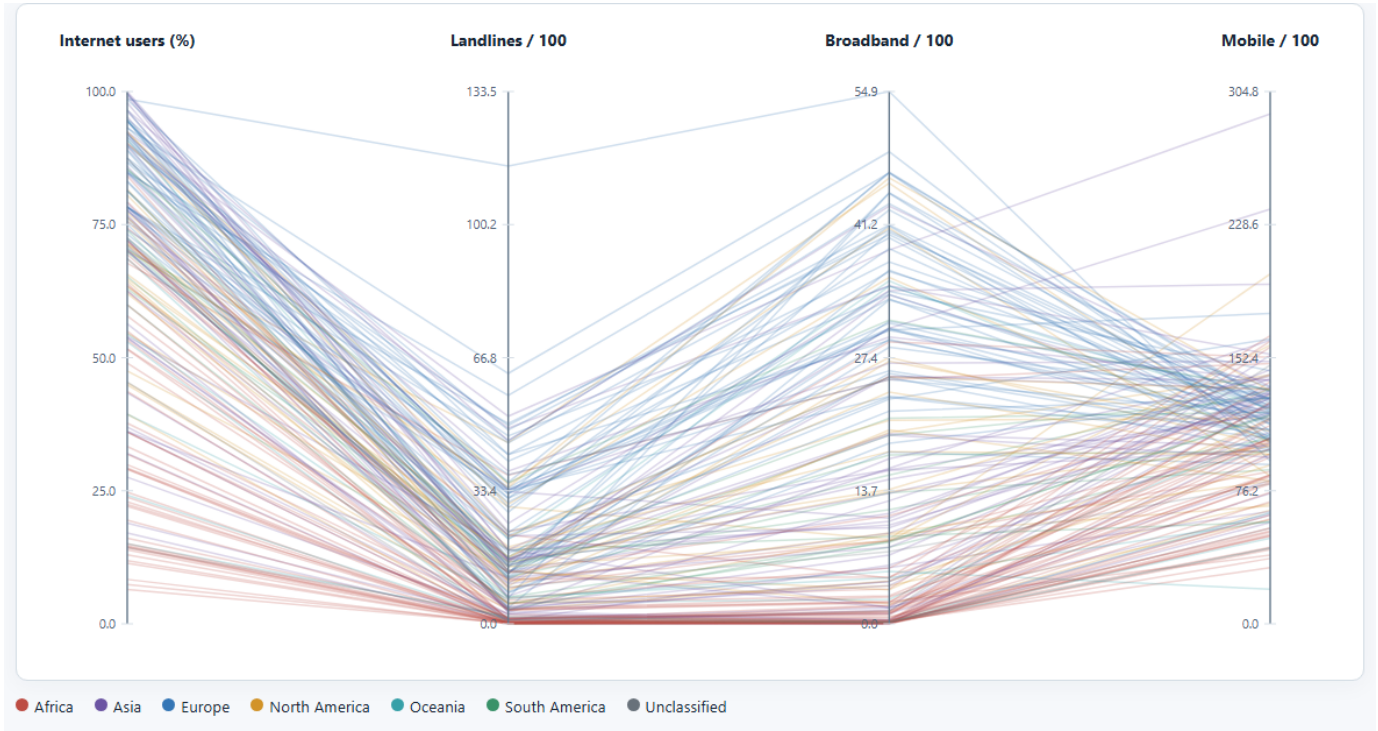


Fig. 4. Straight parallel coordinates for the 2020 matched panel.

A. *Straight parallel coordinates for the 2020 matched panel*

Question (Fig. 4). Which combinations of connectivity measures occur in the same countries?

How to read the graph. Each polyline represents one of the 175 matched countries. Its intersections with the four vertical axes give internet adoption, fixed telephones, broadband and mobile subscriptions in their labeled units. Region determines color. The domains are fixed across the matched 2015 and 2020 comparison, rather than being stretched independently for each year.

Detailed interpretation. Read one polyline across all four measured axes before comparing the whole collection. Low landline intercepts combined with much higher mobile intercepts illustrate why one infrastructure indicator cannot summarize a country. However, a steep segment between differently scaled axes is not a numerical growth rate, and crossings do not show temporal transitions. This is a single-year profile view. The straight-line baseline is retained because it lets the reader check whether later bundling changes readability while preserving the measured values.

Limits. Read values at axis intersections, then follow the same line between neighboring axes. Slopes between axes with different units are not numerical rates of change. Crossings summarize relative ordering, but dense crossings can hide

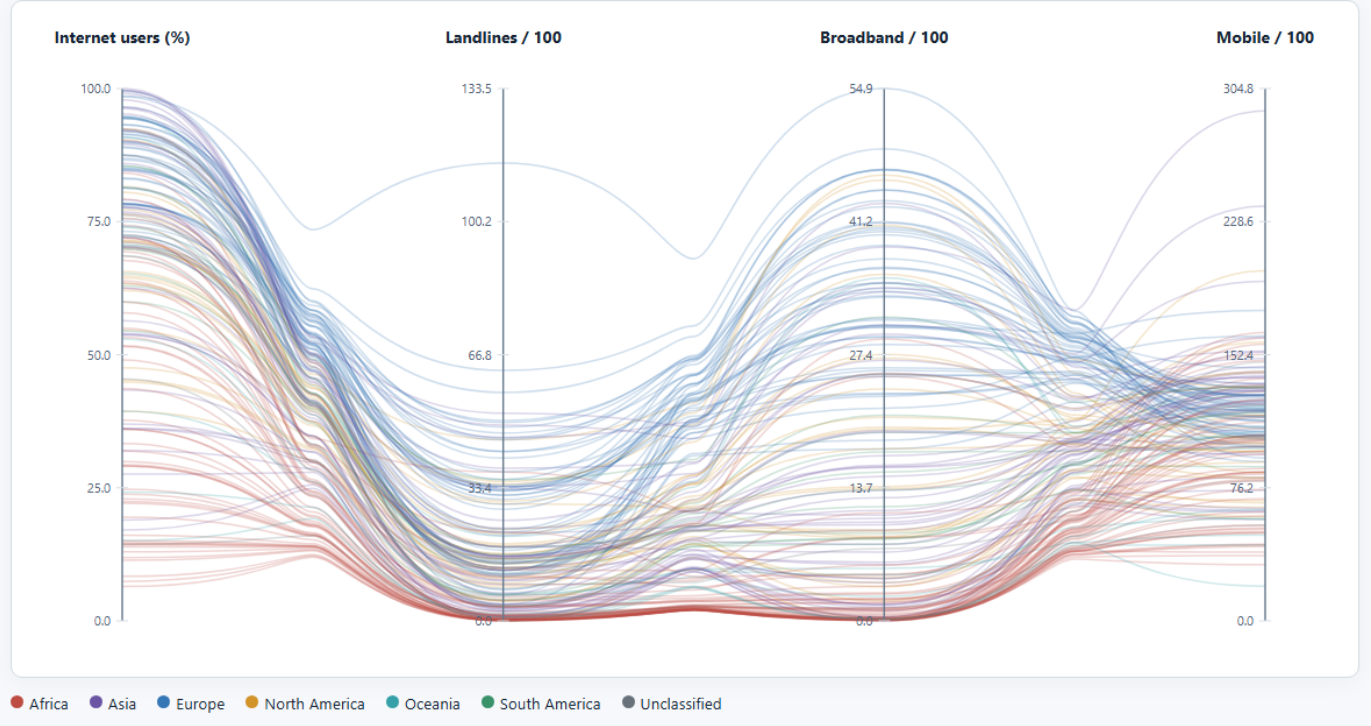


Fig. 5. Region-bundled parallel coordinates for the same countries.

individual countries; brushing or selection is needed for a precise profile comparison.

Takeaway. A country's mobile subscription intensity, fixed infrastructure and adoption need not move together in a 2020 profile. Four explicit axes preserve those differences; subscriptions above 100 remain counts of subscriptions, not people.

B. Region-bundled parallel coordinates for the same countries

Question (Fig. 5). Does moderate bundling help readers follow supplied regional groups?

How to read the graph. The countries, indicator axes, domains and regional colors are identical to Fig. 4. A 50% bundling setting bends the intermediate line segments toward a regional centroid. All measured intersections with the axes are preserved; intermediate curves are constructed drawing paths rather than extra observations.

Detailed interpretation. The intervention here is the 50% region-centroid bundling, not a change in the dataset. Compare the same axis intercepts with the straight baseline: their locations should agree even where the connecting paths bend noticeably. A visually tight regional bundle may result partly from the rendering rule itself. It therefore helps a reader follow known group membership but is not independent evidence that regions form compact statistical clusters. Precision still comes from the intercepts, hover values and brushing, rather than from the curves between axes.

Limits. Compare the two plots at the axis intercepts to confirm that the measurements did not change. A shared curve between axes should not be read as evidence that countries have identical values. Because grouping uses existing region labels, bundle separation cannot be presented as an independently discovered clustering result.

Takeaway. Moderate bundling makes supplied regional paths easier to follow without moving their measured axis values. The resulting bundles visualize the grouping rule; they do not discover or prove regional clusters.

C. Pairwise relationships across four telecom measures

Question (Fig. 6). Which pairwise relationships explain the patterns seen in the four-variable profile plot?

How to read the graph. The lower triangle contains all six pairwise scatterplots of adoption, landlines, broadband and mobile in 2020. Country color gives supplied region. Diagonal panels show equal-country histograms; the upper triangle gives descriptive Spearman coefficients. Axis units remain explicit. The full figure displays four measured variables, while each individual scatter cell is bivariate.

Detailed interpretation. The matrix checks the four-variable scatter's less precise color and area channels using pairwise positions. Landlines and broadband have rho 0.866, while adoption and broadband have rho 0.831; points still vary around both relationships. Mobile shows weaker pooled associations with the fixed indicators. The diagonal histograms reveal skewed distributions that can give a few countries substantial visual influence. Rank correlation reduces dependence on the exact numerical spacing, but it remains a descriptive monotonic summary and does not replace inspecting the scatter for exceptions or regional composition.

Limits. Pooled correlations can reflect differences between regions and need not hold within each region. There are no significance stars or causal claims. Diagonal binning is

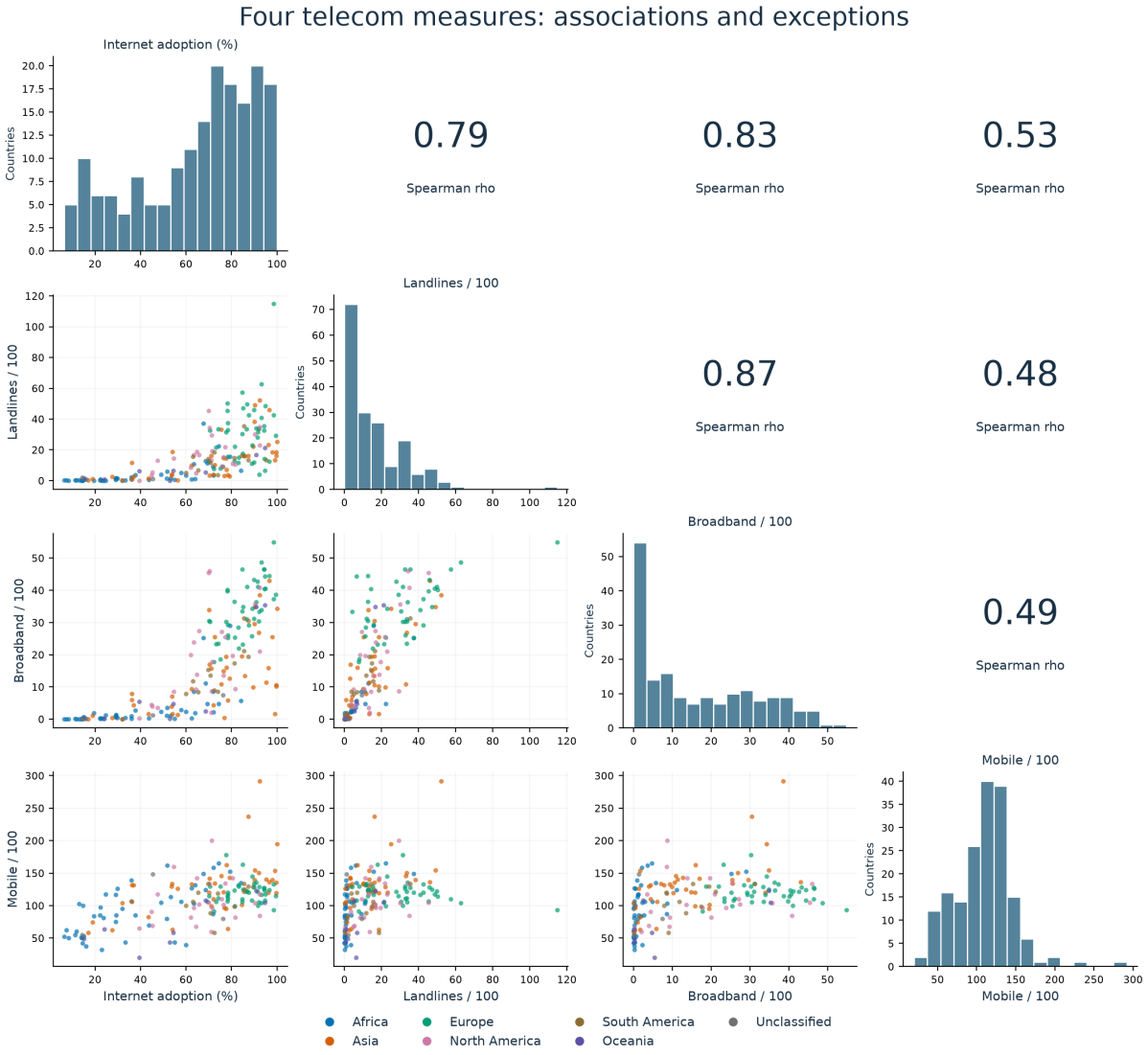


Fig. 6. Pairwise relationships across four telecom measures.

descriptive and should not be treated as evidence of distinct country clusters. Subscription units differ from adoption percentages; high mobile values do not imply universal access.

Takeaway. Fixed infrastructure measures share strong pooled rank associations, but the scatter cells preserve their exceptions and distributional skew. The matrix validates relationships that bubble color and size alone are harder to assess precisely.

D. Matched-country changes reveal divergent infrastructure paths

Question (Fig. 7). Does the strong association of infrastructure levels also describe changes within countries?

How to read the graph. Each country appears once in its supplied-region facet. X gives landline change per 100 and Y adoption change in percentage points, both 2020 minus 2015. A zero-centered diverging scale gives broadband change per 100. Zero lines define the quadrants. Panel headings count positive-adoption, negative-landline cases; points on zero boundaries are excluded from that strict count.

Detailed interpretation. Exactly 126 countries lie in the adoption-gain, landline-decline quadrant. That 72% share is a transparent descriptive count, not a fitted model. The broadband-adoption association falls from rho 0.831 for 2020 levels to 0.054 for 2015-2020 changes on identical countries. Stable cross-country differences, different starting levels and the timing of changes can make these comparisons diverge, but the chart cannot identify which explanation is responsible. It therefore qualifies the level-based story without establishing that broadband growth has no effect.

Limits. Changes compare two endpoints and cannot identify the year of transition. Population changes can also alter subscriptions per 100. Differences in rank correlations across levels and changes do not establish causation, substitution, or statistical significance. Region is context, not an income proxy. A small change correlation does not imply that broadband is irrelevant.

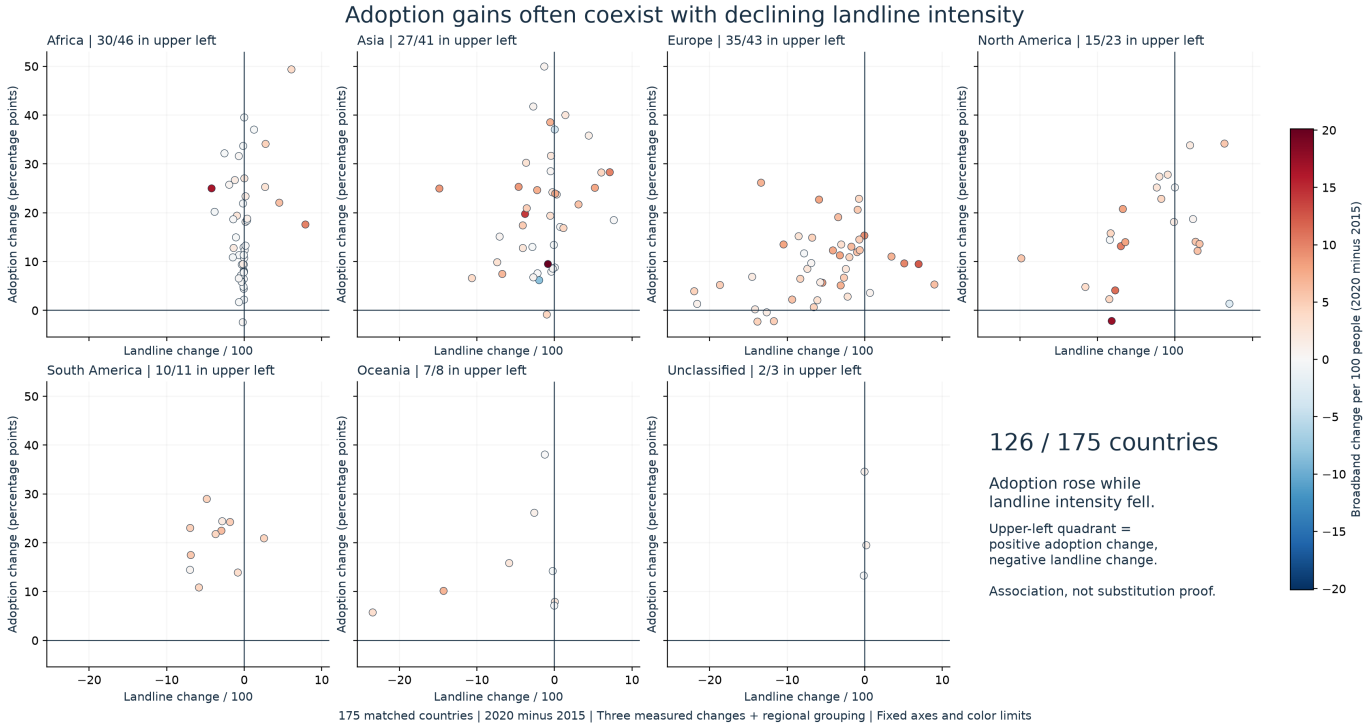


Fig. 7. Matched-country changes reveal divergent infrastructure paths.

Takeaway. Adoption rises while landline intensity falls in 72% of matched countries, yet broadband and adoption changes are only weakly rank-associated. Strong differences between countries must not be read as a causal account of change within countries.

Takeaway. China and India dominate the included Asian user count, but rectangle size does not establish universal adoption. Read area as scale of use and color as prevalence; the two answer different questions.

E. Zoomed region hierarchy for Asia

Question (Fig. 8). Where are included internet users concentrated within Asia?

How to read the graph. A rectangle is a country leaf inside the World-to-region-to-country hierarchy. Rectangle area is proportional to the number of internet users, whereas sequential color represents the percentage of that country's population using the internet. The screenshot is zoomed into Asia; the breadcrumb records the current branch.

Detailed interpretation. China and India together account for more than half of the 2.574 billion included internet users in the Asia view. This concentration explains the large rectangles without requiring either country to have the highest adoption percentage. The color channel answers that separate question. A large population can produce many users even at incomplete adoption, whereas a small country with near-universal adoption can occupy little area. The denominator is the included-country total, so the result should not be generalized to missing countries or treated as a population-weighted average adoption rate.

Limits. Read the parent total first, then inspect both size and color for a country. Large China and India rectangles describe user counts, not land area or universal access. The screenshot's regional total is a subset of the included-country total; neither should be silently substituted for a complete global estimate.

F. Zoomed high-adoption band for Asia

Question (Fig. 9). What is Asia's composition inside the high-adoption group?

How to read the graph. The remodeled hierarchy is World-to-adoption-band-to-region-to-country. Leaves still use internet-user counts for area and adoption share for color. The displayed branch is Asia within the analyst-defined 75-100% band; countries outside that band remain elsewhere in the hierarchy rather than being assigned new values.

Detailed interpretation. This zoom contains 24 high-adoption Asian countries, compared with the 43-country regional view, and a much smaller included user total. The hierarchy first asks which countries belong to the 75-100% band and then where they are located. It does not report users lost after changing the hierarchy. The displayed leaves differ between these two zoom states, so they should not be compared as equal-population experiments. A controlled comparison of the hierarchy algorithms would show the same full leaf universe at the root before zooming.

Limits. Check the breadcrumb and parent total before comparing rectangle sizes. A country may be prominent in Asia's overall user count yet absent from this high-adoption branch. Percentile-like visual groupings should not be inferred: the four adoption bands are fixed percentage thresholds, not equal-sized country groups.

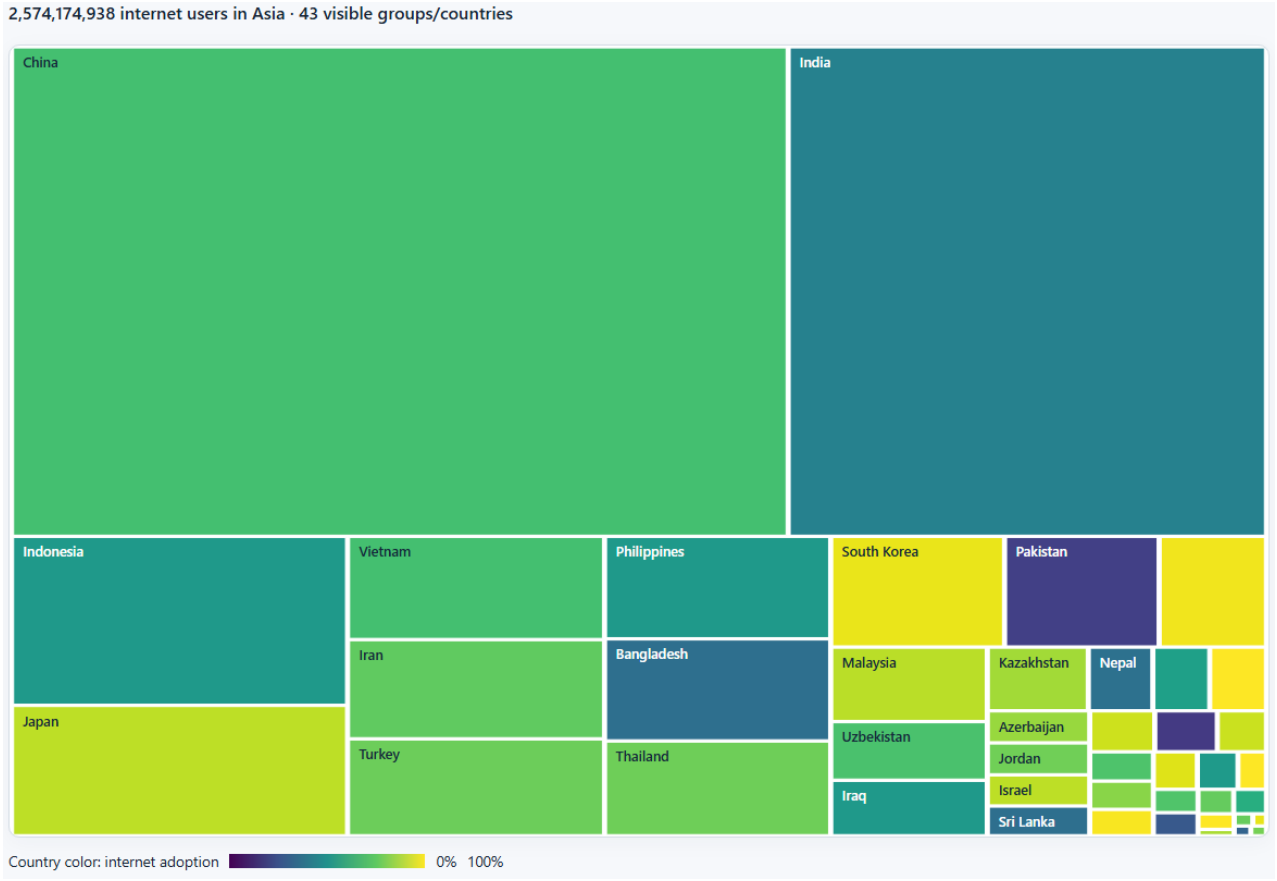


Fig. 8. Zoomed region hierarchy for Asia.

Takeaway. The 497.5 million users in this view belong to a selected high-adoption subset, not a reduced version of Asia's population. Remodeling changes the route of inquiry; the different zoom states also change which countries are displayed.

VI. Country change: compare progress and absolute inequality

A fixed 159-country panel keeps membership constant from 2000 to 2020. Quantile changes distinguish broad progress from absolute convergence. Regional dumbbells restore the compact 2015-2020 comparison. The two required horizon variations retain trajectories and the two ridgeline variations retain distributions.

A. Adoption quantiles and matched-country gains

Question (Fig. 10). Did broad internet-adoption progress also narrow the absolute gap between lower- and higher-adoption countries?

How to read the graph. Panel A compares adoption deciles in 2000 and 2020 on the fixed 159-country complete annual panel. Panel B contrasts differences of corresponding deciles (teal) with deciles of the 159 individual country gains (orange). Each country has equal weight. Values are empirical linear-interpolated quantiles; changes are percentage points.

Detailed interpretation. On the same 159-country panel, the 90th-minus-10th percentile adoption gap widens from 28.63 to 69.84 percentage points. The median of the adoption distribution rises by 69.16 points, whereas the median individual country gain is 62.09 points. Panel A therefore supports broad improvement without absolute convergence on this particular spread measure. Panel B shows why a distributional shift must not be read as the gain of a typical matched country: reordering and heterogeneity affect the two summaries differently. The analysis describes included countries, not a population-weighted distribution of people, and does not identify the causes of progress.

Limits. A country at the 10th percentile in 2000 need not occupy that percentile in 2020. Differences of quantiles are not quantiles of differences. No causal or inferential claim is made and no confidence intervals are shown. The P90-P10 spread measures absolute percentage-point inequality; relative inequality can behave differently. Bounds at zero and 100 and complete-case selection affect interpretation.

Takeaway. Adoption rises across the distribution while the absolute P90-P10 gap widens; broad progress and absolute convergence are different claims.

B. Matched-panel regional dumbbells

Question (Fig. 11). How much do regional adoption medians change on exactly the same country panel?

How to read the graph. Each row is a fixed regional group from the 175 countries complete in both 2015 and 2020. Horizontal

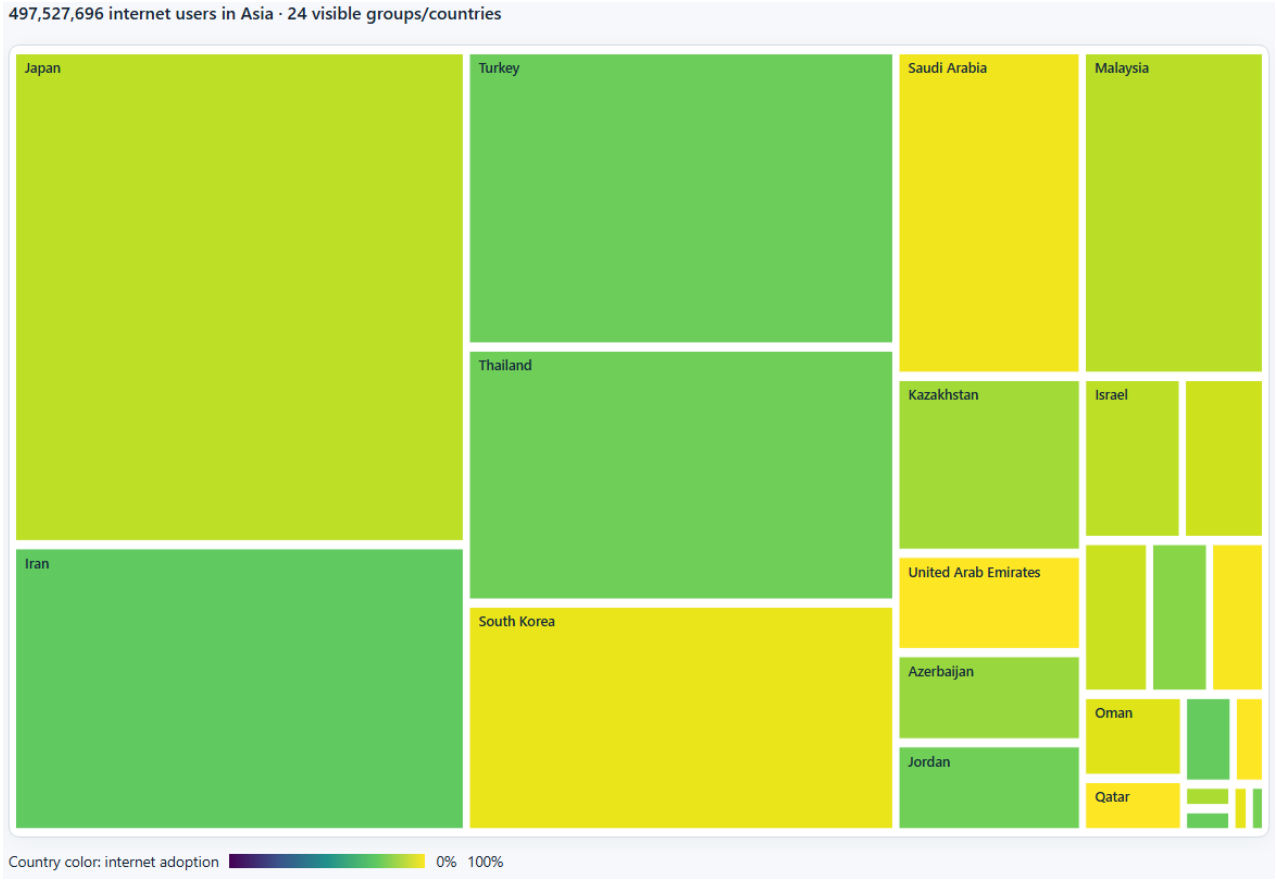


Fig. 9. Zoomed high-adoption band for Asia.

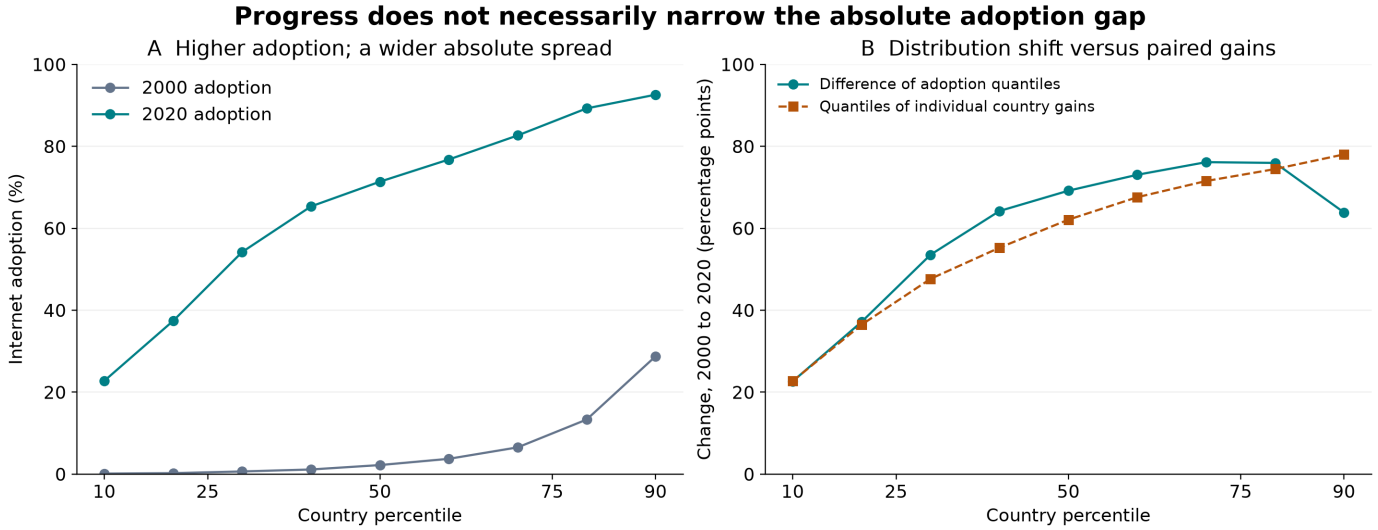


Fig. 10. Adoption quantiles and matched-country gains. Method reference: [1].

position gives median internet adoption percentage. Blue and orange endpoints identify the two years; the connecting segment and annotation give their difference in percentage points. Labels state country counts.

Detailed interpretation. The dumbbell offers an efficient regional summary that the country-change scatter cannot provide at a glance. Asia's median shifts by 26.55 percentage points, the largest increase among classified regions on this matched panel. A movement in the regional median does not mean each country moved by that amount or even moved upward. The 175-country panel also differs from the 159-country long-run panel, so small differences from the horizon medians need not indicate an error. Each plot answers a different coverage and time-span question.

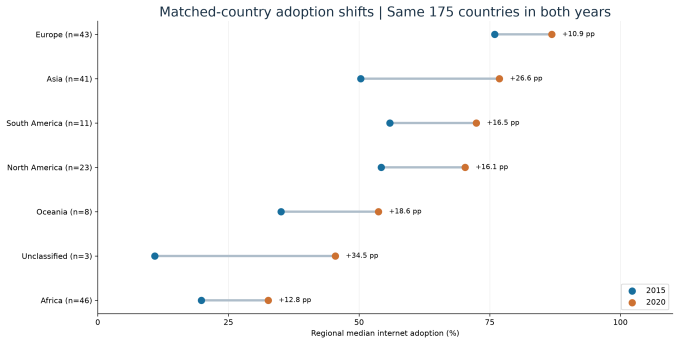


Fig. 11. Matched-panel regional dumbbells.

Limits. Compare each row's endpoints, then inspect the labeled difference. The statistic is the difference of two group medians, not the median of paired individual-country changes. Unclassified groups retain missing region labels without pretending they form a geographical region. Equal-country weighting does not estimate population-weighted regional adoption.

Takeaway. Asia has the largest classified-region median shift on the 175-country matched panel, at 26.55 percentage points. Retain this concise group summary alongside country-level changes; it does not describe every country's trajectory.

C. Country horizon plots

Question (Fig. 12). How do selected countries approach or exceed majority internet use over time?
How to read the graph. Each row follows one selected country's annual adoption from 2000 to 2020. The plotted quantity is adoption minus 50 percentage points. Two 25-point bands per side are mirrored and folded: red denotes below the reference and blue above it, with darker bands indicating greater distance. Time runs left to right.
Detailed interpretation. The country rows are a deliberately selected set of low/high 2020 examples within regions. Their contrast is useful for showing heterogeneity that a regional median can hide, but it is not an estimate of how common each trajectory is. Follow one row horizontally: changing saturation indicates distance from the 50% baseline, while the sign color indicates its side. A row may become less dark red as adoption rises without becoming blue. The construction example later explains that apparent paradox using the original values.
Limits. Read hue as the sign relative to 50%, saturation as distance, and changing shape as temporal movement. A red row can represent rising adoption that remains below 50%; red does not mean decline. For precise adoption values, return to the annual table or the unfurled construction in the documented animation samples.

Takeaway. Selected countries illustrate contrasting routes toward majority internet use. The selection demonstrates variation rather than its prevalence, and a red horizon can still represent improving adoption below 50%.

D. Regional median horizon plots

Question (Fig. 13). Do regional median trajectories cross the majority-use milestone at similar times?
How to read the graph. Each row shows an annual median over a fixed set of countries in one classified region. Every included country has equal weight. The sign, two 25-point bands and 50% baseline match Fig. 12, so a blue interval indicates that the regional country median exceeds majority use.
Detailed interpretation. Europe's 2020 included-country median of 86.86% exceeds Africa's 31.89% by about 55 percentage points. Reading that endpoint together with the trajectories distinguishes the timing of crossing the 50% milestone from the final level reached. The fixed 159-country panel prevents changing availability from explaining the regional time paths; six classified regions contain 158 of those countries. Equal-country medians still hide within-region variation, so the country horizons and regional ridgelines are necessary companions rather than interchangeable repetitions.
Limits. Compare the timing of color changes, then inspect the reported 2020 medians. The regional statistic is not the share of all regional people who are online: large and small countries receive equal weight. A median below 50% also does not mean every included country in the region is below that threshold.

Takeaway. On a fixed country panel, regional progress is uneven: Europe crosses majority use earlier, while Africa's median is still 31.89% in 2020. Regional improvement does not establish that the gap has disappeared.

E. Whole-panel ridgeline variation

Question (Fig. 14). How does the whole fixed panel's adoption distribution shift between selected years?
How to read the graph. Horizontal position is adoption percentage on the common 0-100% scale. Each ridge estimates the distribution across the same 159 countries in one year using a reflected Gaussian kernel with bandwidth 6 percentage points. Ridge heights are separately normalized; ticks mark raw medians and year labels identify the slices.
Detailed interpretation. The median shifts from 2.20% to 71.37%, a rise of 69.17 percentage points, while the distribution remains spread across adoption levels. These are two different findings: a typical included country has progressed substantially, yet the entire population of countries has not collapsed into one narrow band. Keeping the same 159 countries means this change cannot be attributed to richer-data countries entering later years. Density peaks depend on smoothing, so the robust conclusions concern the broad shift and persistent spread, not the exact number of bumps.
Limits. Compare where density is concentrated and how broadly each ridge extends, rather than comparing peak heights as counts. A higher median can coexist with countries at both ends of the scale. Small density bumps depend on smoothing and should be checked using the documented animation samples before being described as distinct adoption groups.

Takeaway. The median country's adoption rises by 69.17 percentage points from 2000 to 2020, but the broad final distribution still contains unequal outcomes. A higher center is not the same as convergence.

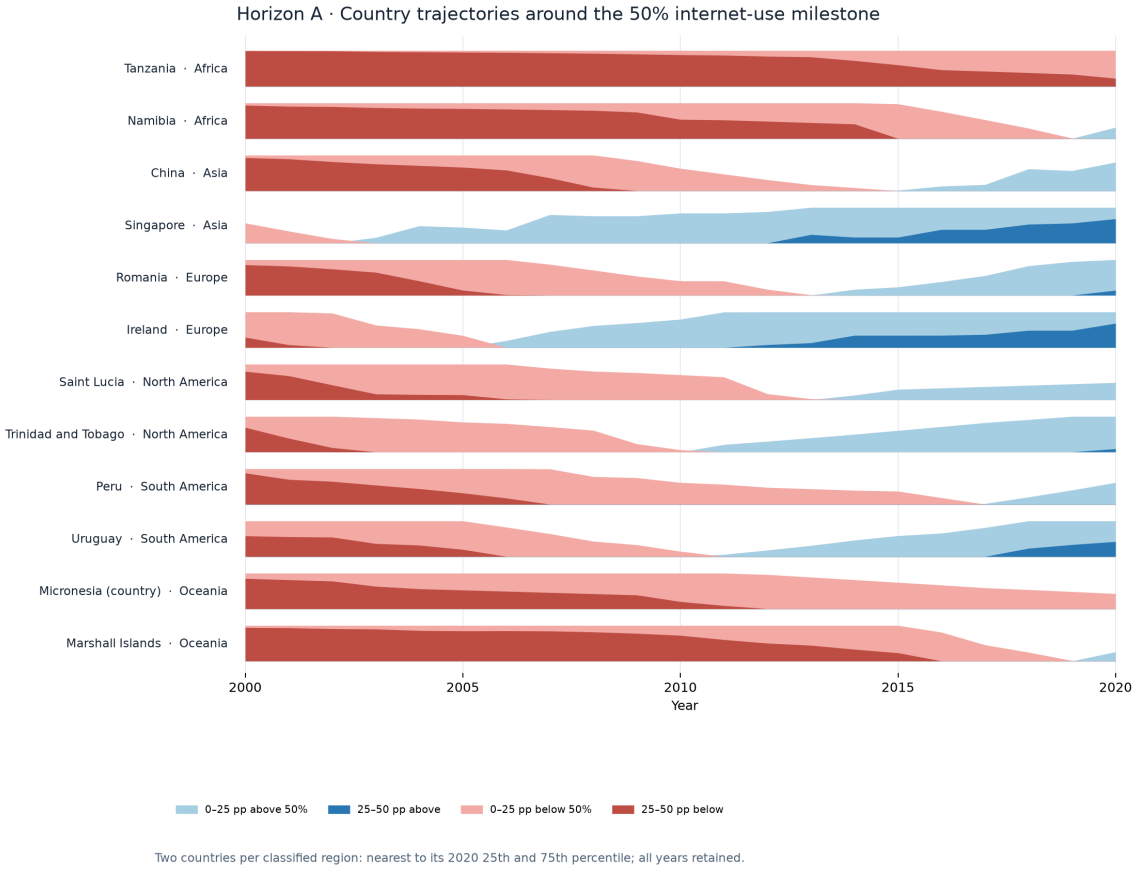


Fig. 12. Country horizon plots. Method reference: [9].

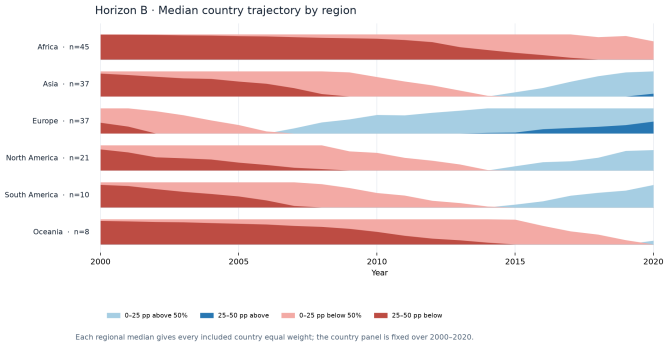


Fig. 13. Regional median horizon plots.

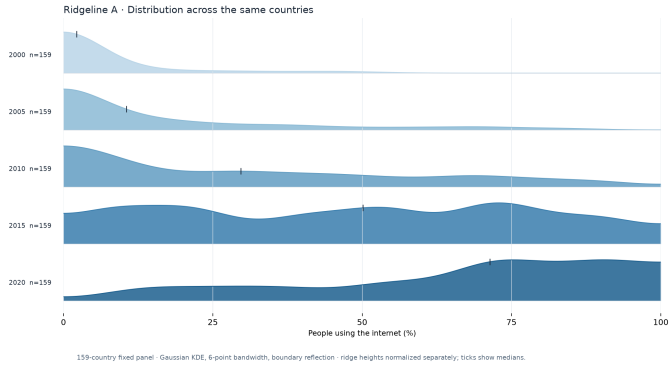


Fig. 14. Whole-panel ridgeline variation.

F. Region-faceted ridgeline variation

Question (Fig. 15). Which regional distribution shifts are hidden by the global ridgeline?

How to read the graph. Each facet is a classified region, and each ridge is one of the same five annual slices. The adoption axis, kernel bandwidth, boundary treatment and median ticks match Fig. 14. Country counts are fixed within region; ridge height remains independently normalized rather than representing the size of the region.

Detailed interpretation. Faceting asks whether the pooled shift reflects comparable movement inside every region. Inspect the horizontal locations and spreads of ridges at matched years, rather than comparing peak heights as counts. The small Oceania and South America groups contribute only eight and ten countries, making their smooth curves particularly sensitive to individual observations. A region with a compact high-adoption distribution and one with a broad lower distribution can be concealed by a single global ridge. These panels explain composition, not independent statistical clusters.

Limits. Compare horizontal shifts within each facet and differences between regional medians. Do not compare filled area as regional population or numbers of users. Oceania and South America have particularly small country panels, so peaks there are fragile descriptive shapes; regional membership and

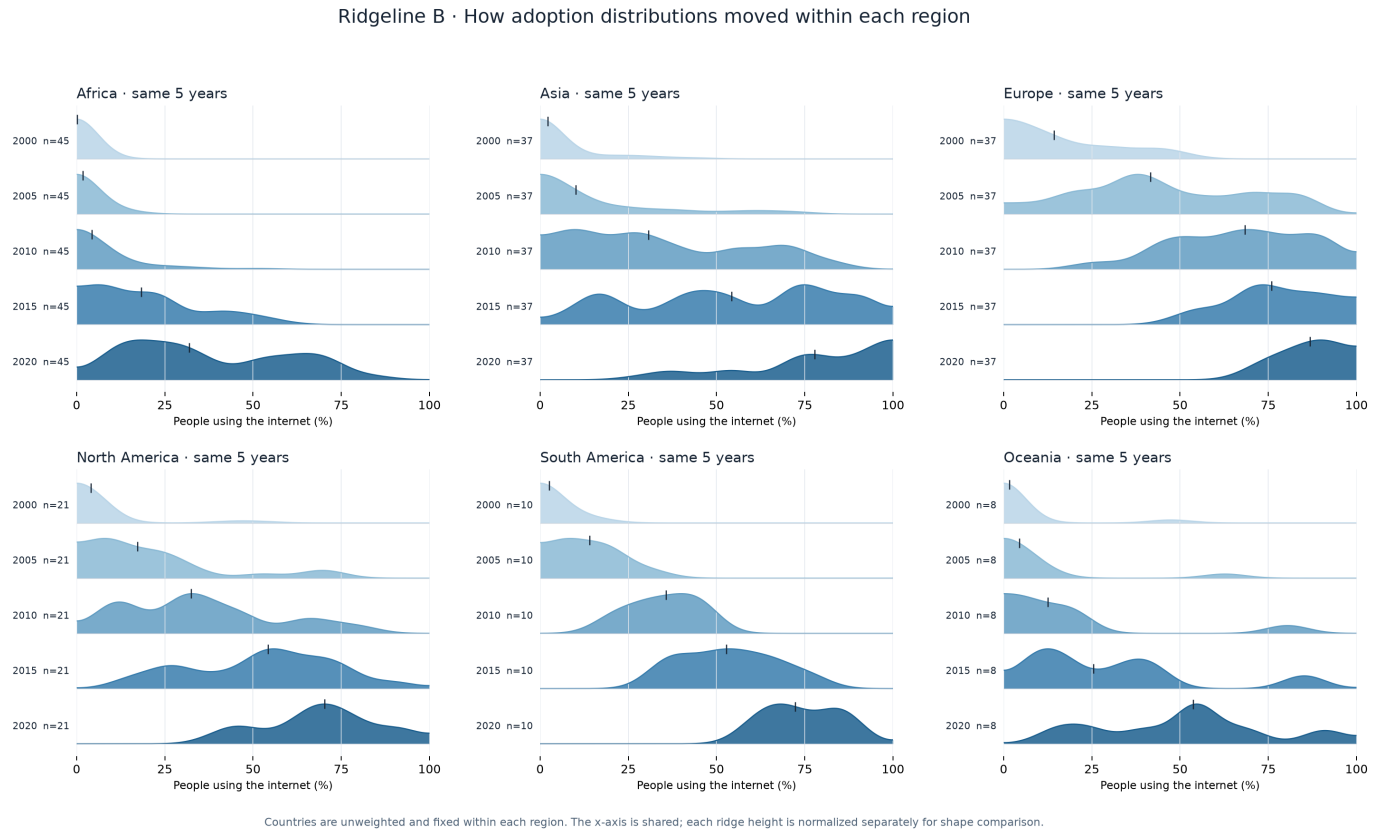


Fig. 15. Region-faceted ridgeline variation.

missing-country coverage also limit generalization.

Takeaway. Regional distributions show which groups contribute to the global adoption shift and which retain broad variation. Apparent peaks in the smallest regions are weak evidence unless supported by raw values and smoothing checks.

VII. Scalar fields: compare visual choices on identical weather inputs

The SciVis investigation uses a separate dataset. Controlled scalar experiments first show what color mapping, temporal normalization and contour spacing change. The measured weather fields remain fixed within each comparison; apparent contrast must not be mistaken for changed temperature or rainfall.

A. Continuous, discrete, linear and logarithmic mapping

Question (Fig. 16). How do palette classes and normalization alter the visibility of the same scalar fields?

How to read the graph. The temperature pair compares a continuous sequential palette with fixed 5-degree classes. The rain pair compares linear and logarithmic positive-rain mapping for the same date. Gray is missing, white is dry, and the stated low-positive-rain category is distinct from zero. Color bars define units and limits, including saturation.

Detailed interpretation. With 68.7% of valid cells dry, the dominant rainfall distinction is zero versus positive precipitation before any subtle positive-value contrast. Logarithmic mapping exposes differences among smaller positive amounts, while a linear scale emphasizes the larger values. The 1.01% of cells above the displayed rain limit cannot be ranked visually after saturation. The temperature comparison changes both palette and classing, so it demonstrates their combined effect rather than isolating discretization as a single cause. This limitation should accompany any claim about which rendering is clearer.

Limits. Compare identical locations within each variable's pair, not temperature colors against rainfall colors. Saturated rain cells cannot be ranked precisely from their shared endpoint color. The log experiment must not treat zero as a small positive amount; the separately encoded dry fraction prevents that mathematical and interpretive error.

Takeaway. Rainfall normalization changes which positive amounts are distinguishable, while most cells remain dry. Preserve zero and saturation information; the temperature pair shows a combined palette/classing change rather than an isolated discretization experiment.

B. Global versus local extrema

Question (Fig. 17). Is stronger within-day contrast compatible with reliable between-day color comparison?

How to read the graph. The two columns compare one shared scale against a scale fitted to each displayed day's actual minimum and maximum. Each row uses the same date and field in both columns. The local color bars change by day; the shared limits are computed over the original eight-field sampled cube, not over the full 92-day cache.

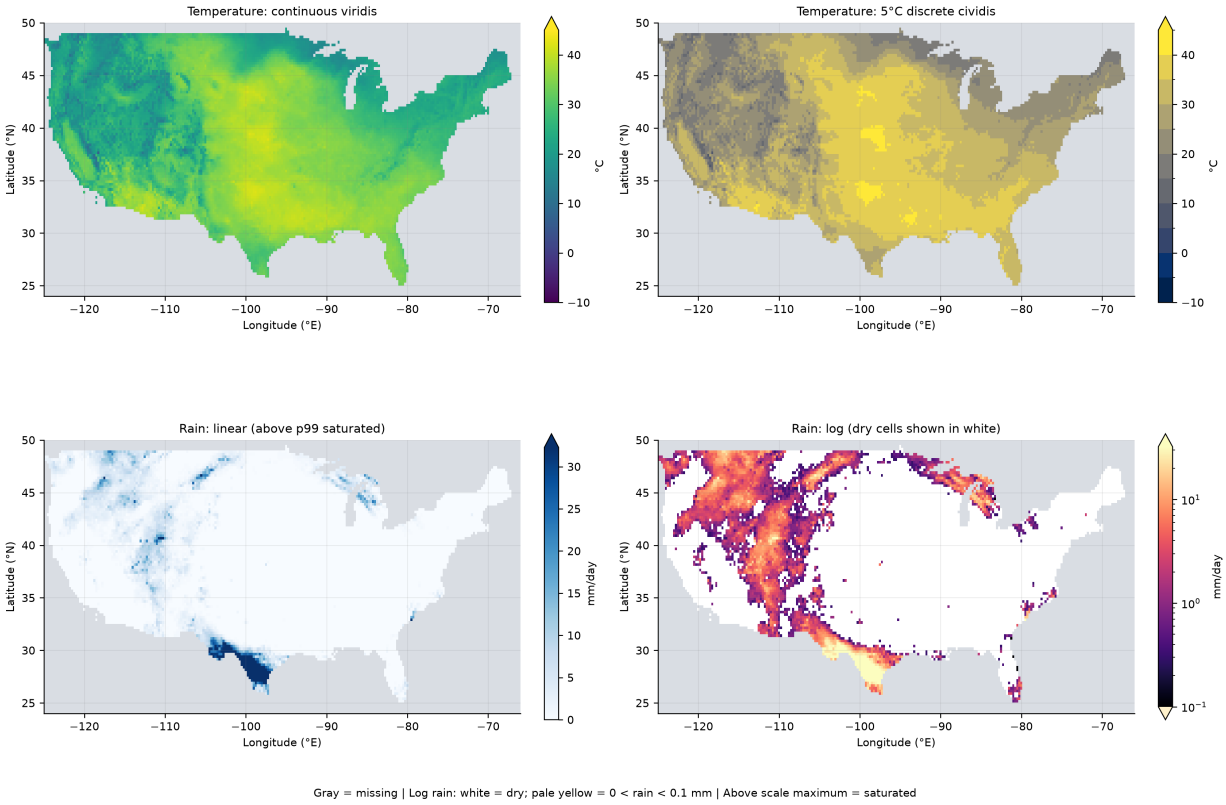


Fig. 16. Continuous, discrete, linear and logarithmic mapping. Method reference: [10].

Detailed interpretation. The local scale stretches each day's range across the palette, making within-day variation more visible even on a relatively uniform day. That same benefit can make a cooler day look as visually warm as a hotter one. The narrowest and widest ranges quoted in the caption are descriptive of the sampled dates, not climate-normal variability. Shared extrema are preferable for comparing absolute temperatures across dates; local extrema answer a different question about spatial contrast within each date. Neither choice changes the observations.

Limits. Compare spatial structure within a row, then inspect the bars before comparing colors between rows. A yellow local cell on a cooler date may have a lower temperature than a green cell on a warmer date. Changes in apparent intensity therefore cannot be used alone to establish warming or cooling.

Takeaway. Use the shared scale to compare temperatures between dates and the local scale to inspect contrast within a date. Similar colors in locally normalized maps are not evidence of similar absolute temperatures.

C. Contour spacing and fill

Question (Fig. 18). How do contour interval and filling change the readability of temperature boundaries?

How to read the graph. All four panels use the same daily maximum-temperature field. Columns change the contour interval from 5 to 10 degrees; rows change line contours to filled intervals. Geographic axes remain fixed. A contour line traces interpolated locations of equal temperature, while a filled band marks values between neighboring levels.

Detailed interpretation. The 5-degree contours offer more threshold crossings to follow than the 10-degree contours, but also introduce more lines and labels that can obscure one another. Filled bands change the task from tracing an exact boundary to comparing areas inside temperature intervals. Compare the same location in all four panels to attribute differences to interval or fill. The contours interpolate crossings between sampled cells; a smooth line is not evidence that the original field was measured at that finer resolution. This controlled comparison is required methodological evidence.

Limits. Read a boundary through its level or adjacent color intervals, rather than assuming every line represents an independent observation. The Marching Squares family traces transitions between available samples; smooth-looking boundaries do not recover omitted native cells. Identical sampling and extent make differences between panels drawing effects.

Takeaway. Contour interval and filling determine whether boundaries or broad regions are easier to inspect. Finer contours add display detail, not new measurements, and should be chosen for the question rather than visual complexity.

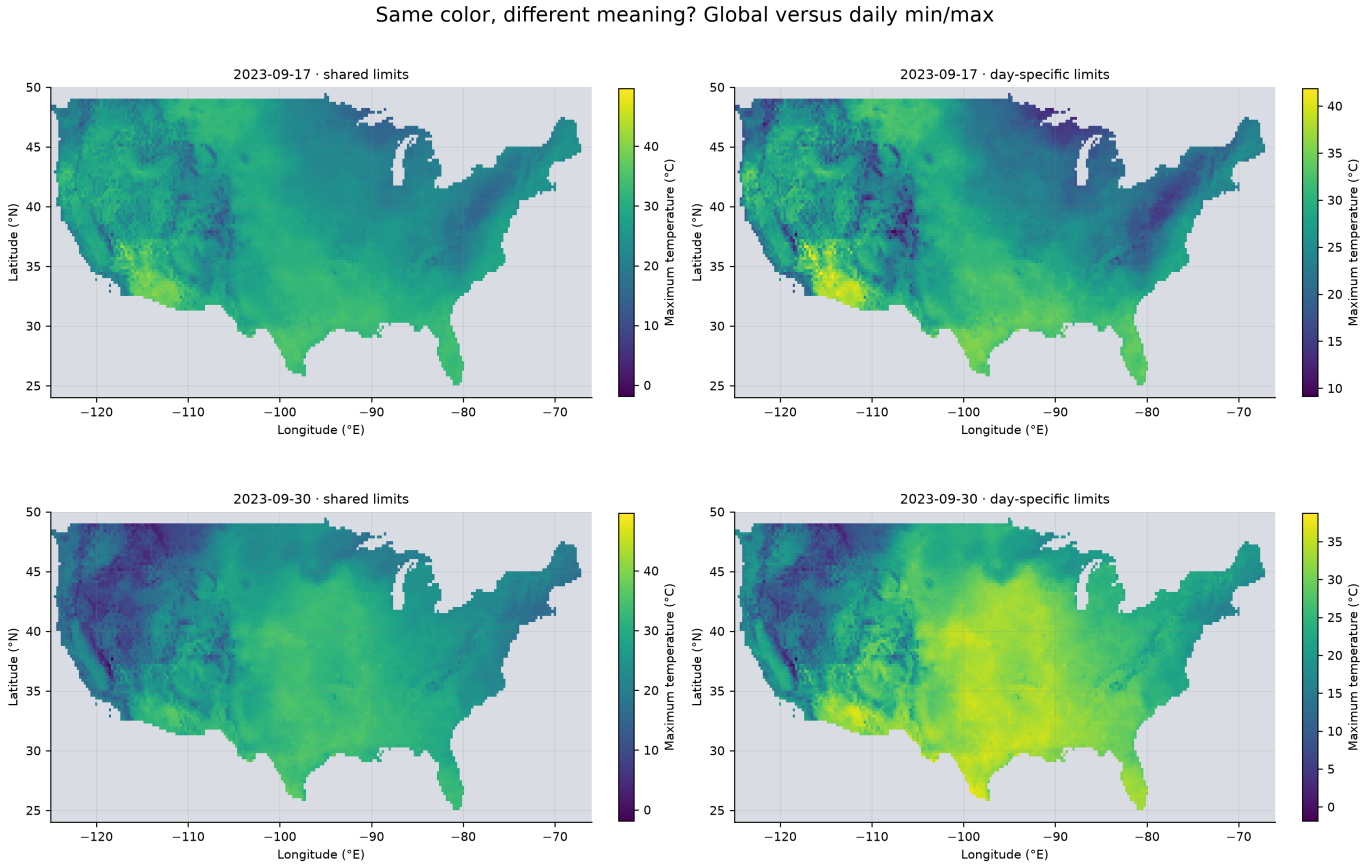


Fig. 17. Global versus local extrema.

VIII. Weather Change: Replace Isolated Snapshots with Complete Daily Evidence

Complete-month differences, daily spread, rainfall totals, heat frequency and time-longitude fields answer different temporal questions. All-period analyses use the 92 daily fields; animation-scale experiments still use only the eight documented sample dates. Each denominator and geographic support is stated.

A. Monthly means with fixed isolines

Question (Fig. 19). How does each month's mean daily maximum temperature vary across the sampled area?
How to read the graph. Each map averages every available daily maximum-temperature field at a cell within that month: 31, 31 or 30 daily values. All maps share one Celsius scale and the same 25- and 35-degree contour levels. Gray denotes missing coverage. These are monthly averages of daily maxima, not averages of all hourly temperatures.
Detailed interpretation. The area-weighted monthly means decrease from 31.19 to 30.16 to 26.55 C, but each panel preserves where warmer and cooler regions occur. Fixed 25 and 35 C isolines provide recognizable thresholds on top of the shared color scale. The result complements, rather than duplicates, the difference map: these panels show absolute monthly levels, whereas the difference map shows the amount of change. The mean of daily maxima must not be paraphrased as the mean temperature across all hours of a month.
Limits. Compare the same locations and the extent of the fixed isotherms between months. A high monthly mean maximum does not specify how many individual days crossed a threshold; Fig. 23 answers that frequency question. Without a multiyear normal, these maps also do not establish whether the months were unusually hot.

Takeaway. Monthly mean daily maxima decline from 31.19 C in July to 26.55 C in September, with spatial structure still visible. Absolute-level maps locate the remaining warmer regions that a change-only map cannot identify.

B. Complete-month temperature differences and spatial distributions

Question (Fig. 20). How does the temperature change across complete months differ from an endpoint-only comparison?
How to read the graph. A zero-centered diverging map encodes September mean daily maximum temperature minus July mean daily maximum temperature at each cell. Blue denotes cooling and red warming. The adjacent weighted empirical distributions show the fraction of each fixed longitude sector at or below a given difference. Both computations use common complete cells and cosine-latitude weights; longitude bands are analyst-defined.
Detailed interpretation. Using all daily fields gives a mean September-minus-July Tmax difference of -4.64 C and cooling over 99.66% of the weighted footprint. The former two-date

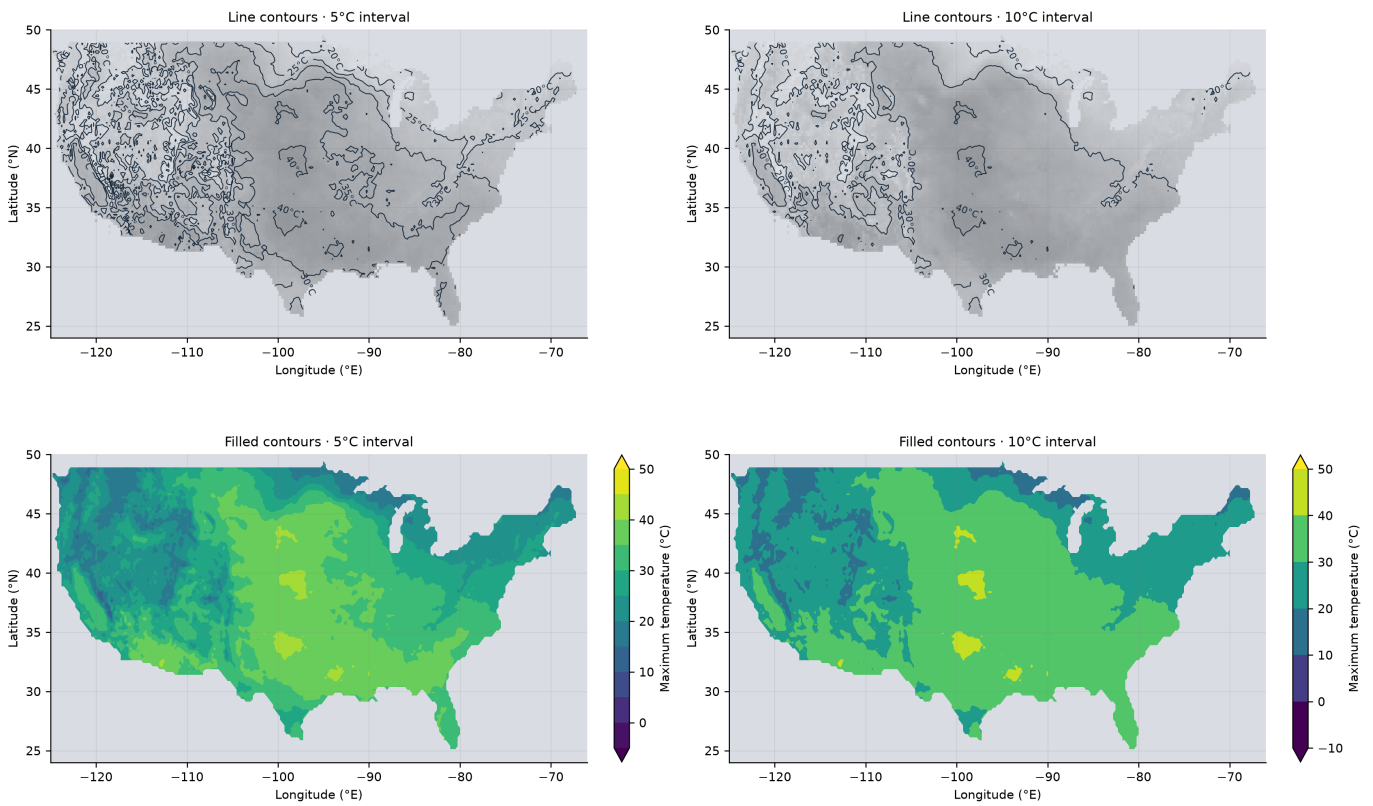


Fig. 18. Contour spacing and fill. Method reference: [11].

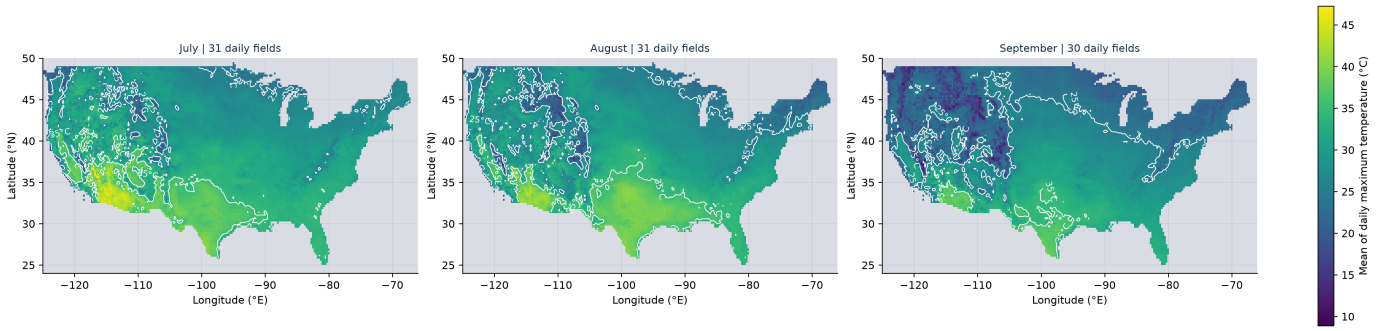


Fig. 19. Monthly means with fixed isolines.

view disagrees in sign over 18.26% of area, which demonstrates the sensitivity of an apparent change to temporal selection. Sector ECDFs then show that cooling is not spatially uniform even where its sign is consistent. This figure does not reject daily snapshots; it limits what a snapshot comparison can answer when the stated question concerns complete months.

Limits. These are differences in means of daily maxima, not differences in mean air temperature. The ECDF is a distribution over space, not a confidence interval. A three-month interval provides no climatological baseline and cannot establish climate change. Subsampling approximates the spatial footprint. Exact zero differences are neither warming nor cooling.

Takeaway. Complete-month means show cooling across 99.66% of sampled area, yet an endpoint-only map disagrees in sign over 18.26%. Temporal aggregation changes the supported conclusion; this within-2023 difference is not a climate anomaly.

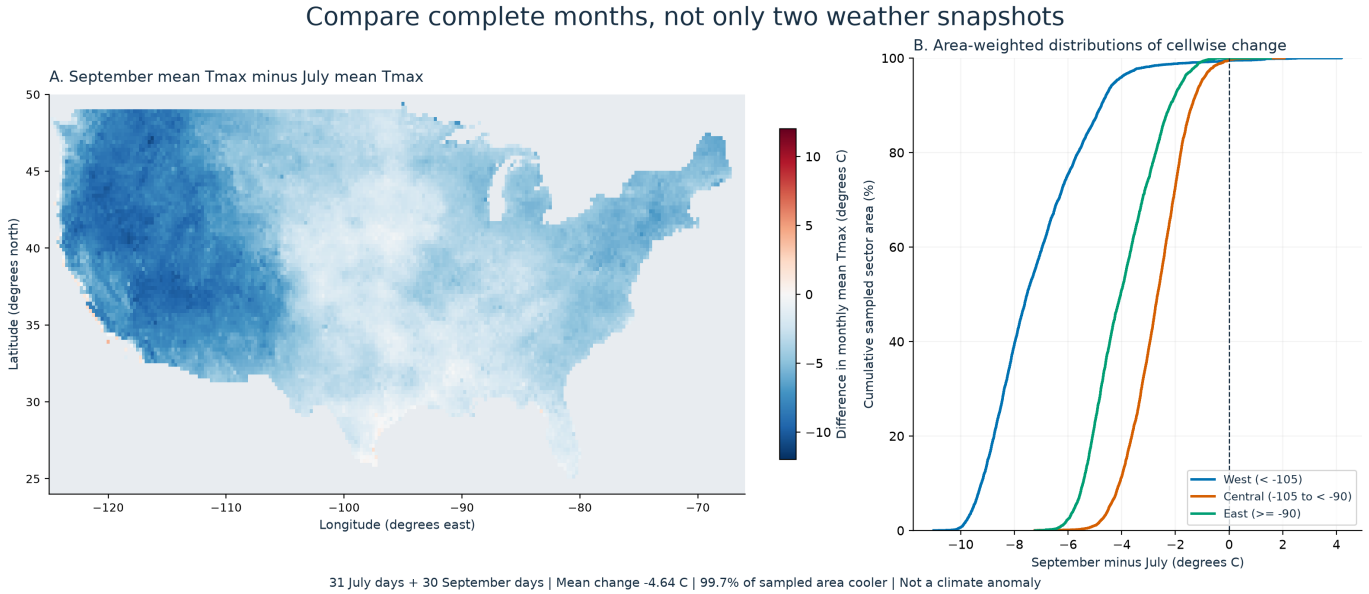


Fig. 20. Complete-month temperature differences and spatial distributions.

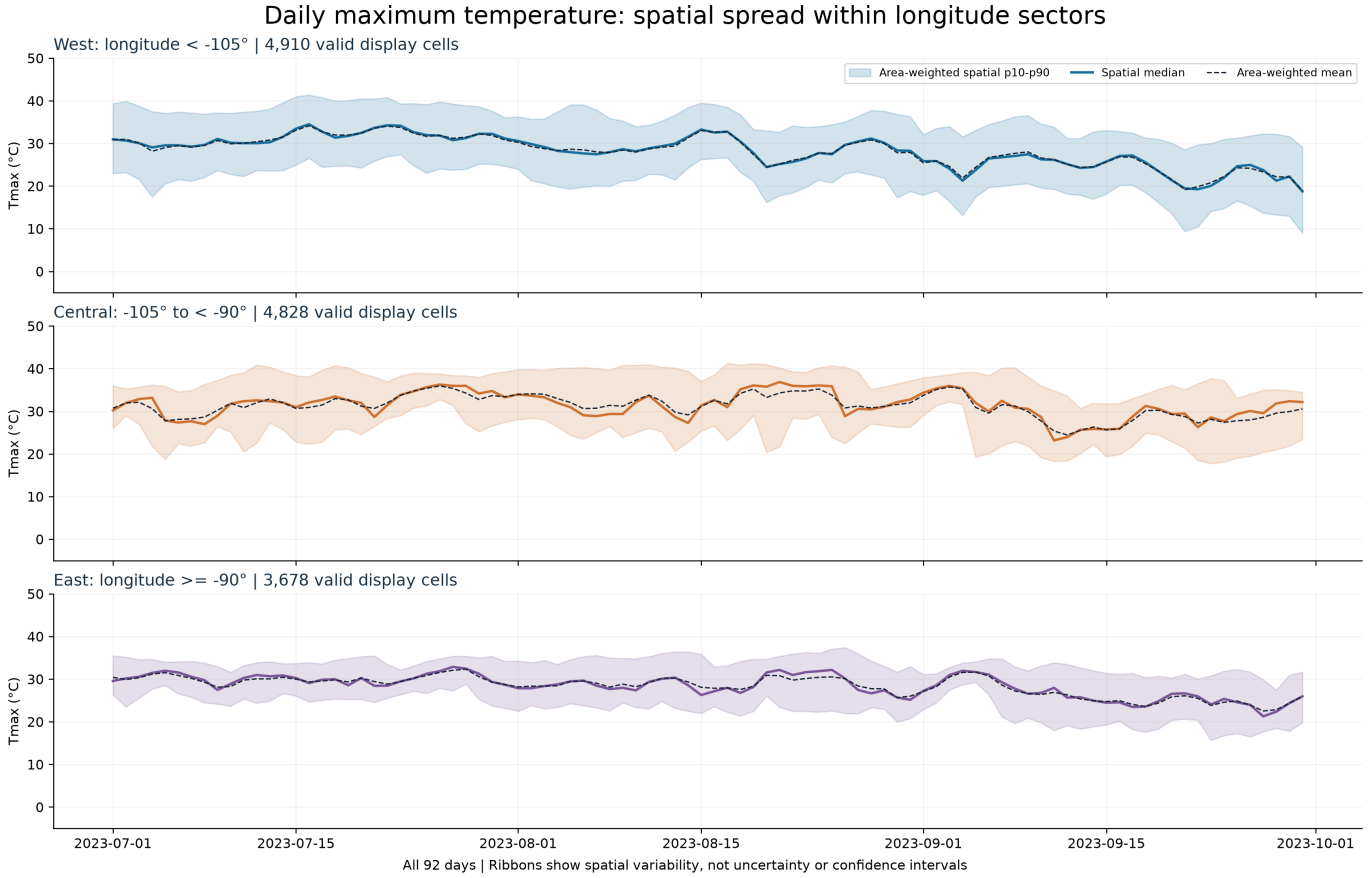


Fig. 21. Daily spatial quantile ribbons.

C. *Daily spatial quantile ribbons*

Question (Fig. 21). How much daily temperature variation and within-sector spatial spread do snapshots omit?

How to read the graph. Time gives all 92 dates. Each facet is a declared longitude sector, with a shared temperature axis. The mean and median summarize daily maximum temperatures using approximate area weights; the ribbon spans the spatial 10th-90th percentiles within the same fixed valid footprint. The sectors are analyst-defined longitude bands.

Detailed interpretation. The western, central and eastern sectors

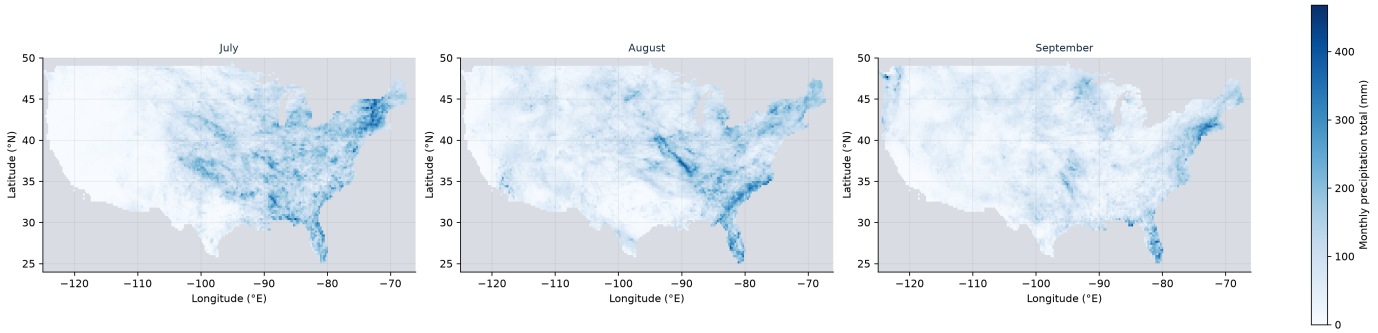


Fig. 22. Complete-month precipitation totals.

summarize different collections of cells, so the ribbons describe spatial spread inside each sector. A wide ribbon means that locations differ on that day; it does not mean the mean was estimated with poor statistical confidence. The central sector's endpoint change is only -0.15 C, but the path between those endpoints still contains substantial movement. This is why the full trajectory contributes information beyond the monthly difference: a small net change can conceal large intervening fluctuations.

Limits. Read a daily central value together with its ribbon width. A broad ribbon indicates varied temperatures across that sector that day, not uncertainty in an estimated mean or a confidence interval. Mean-median separation can indicate asymmetry, but the chart alone cannot attribute weather changes to a particular physical mechanism.

Takeaway. A near-zero central-sector endpoint change does not imply stable daily temperatures. The 92-day paths expose intervening fluctuations, while the ribbons describe differences between locations rather than uncertainty in an estimated mean.

D. Complete-month precipitation totals

Question (Fig. 22). Where does total accumulated rainfall differ between the three complete months?

How to read the graph. Each cell is the sum of its daily precipitation accumulations for all dates of that month. The common sequential scale is in millimeters and uses the actual monthly maximum without clipping. Any cell lacking complete monthly coverage is masked, rather than treating a missing daily value as zero rainfall.

Detailed interpretation. July and August have similar area-mean accumulation, 75.26 and 75.82 mm, while September is lower at 57.08 mm. A similar domain mean does not imply the rain fell in the same places, which is why the maps are retained. September also has one fewer day: these are actual monthly totals, so a claim specifically about daily intensity should instead compare day-normalized means. Masking incomplete months prevents missing observations from being silently interpreted as dry days. The eight animation snapshots cannot reproduce these sums.

Limits. Read both the magnitude and distribution of totals. A monthly sum cannot show whether rainfall came from many light days or a few heavy days; the supporting rainfall distribution retained in the archive examines the daily amount distribution. These are actual month totals, so September's 30-day duration should be considered when comparing it with the 31-day months.

Takeaway. July and August have similar area-mean rainfall totals, while September is lower; the maps reveal their different spatial allocation. Interpret these as complete-month accumulation, with 31/31/30-day durations explicitly retained.

E. Threshold-frequency maps

Question (Fig. 23). How often does daily maximum temperature reach the declared 35-degree threshold?

How to read the graph. At each complete-coverage cell, qualifying days are counted and divided by that month's 31, 31 or 30 days. Maps share a 0-100% sequential frequency scale. Approximate area-day percentages summarize the footprint separately. The threshold is a descriptive analytical choice, not an official heatwave or exposure classification.

Detailed interpretation. Threshold frequency answers how often a location experiences a daily maximum of at least 35 C, rather than how warm its average day is. The area-day fraction falls from 22.8% in July and 22.2% in August to 8.9% in September. Two cells can have a similar monthly mean but different numbers of threshold exceedances. This figure remains a useful temperature-only baseline for the later compound analysis: adding a low-rainfall condition narrows the question and cannot increase the joint frequency above the hot-day frequency on identical support.

Limits. Interpret 50% as half the included days crossing the threshold at that cell, not half its population experiencing heat. The plot does not encode consecutive hot-day runs or nighttime conditions. Neither temporal persistence nor health risk can be inferred from this one daily-maximum threshold alone.

Takeaway. The >=35 C area-day fraction is much lower in September, but this measures frequency alone. It supplies the temperature-only baseline needed to interpret the stricter hot-and-dry condition and its persistence.

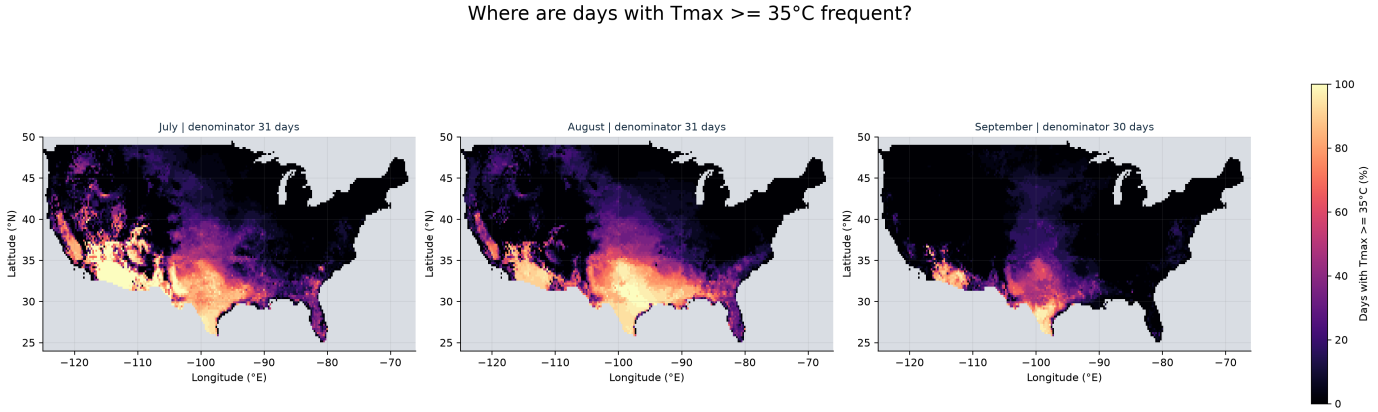


Fig. 23. Threshold-frequency maps.

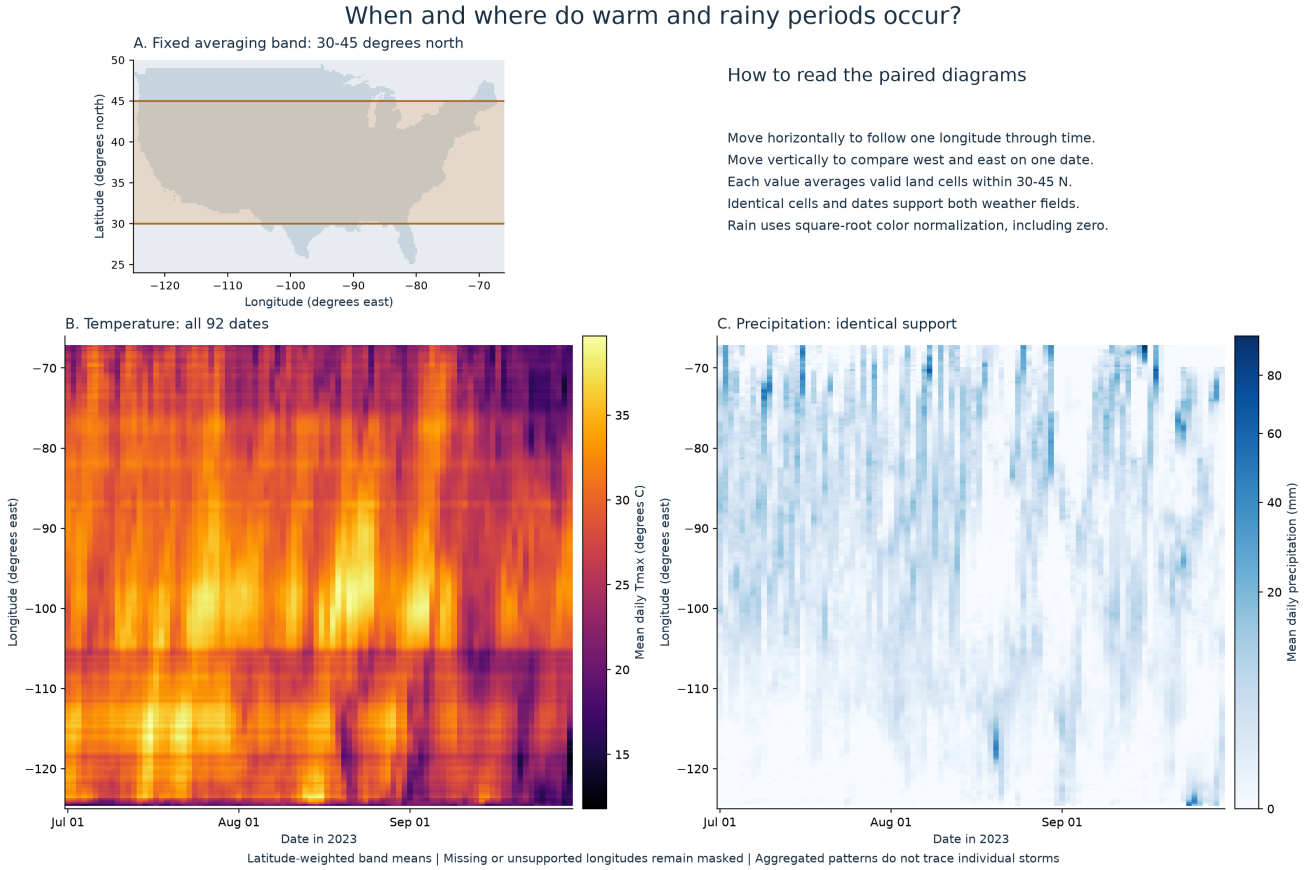


Fig. 24. Paired time-longitude diagrams retain all daily weather fields. Method reference: [14].

F. Paired time-longitude diagrams retain all daily weather fields

Question (Fig. 24). What daily time-space structure is hidden by calendars, sector averages and monthly maps?

How to read the graph. Horizontal position is date and vertical position is longitude in both lower panels. Color shows weighted temperature or daily precipitation in separate aligned panels. The reference map identifies the fixed 30-45 degree north averaging band. Temperature uses a linear scale; rain uses square-root normalization with a colorbar in actual millimeters, including zero. Each longitude retains identical common valid land cells across all dates.

Detailed interpretation. Reading across a longitude follows its entire 92-day band-average trajectory, whereas reading vertically compares locations at the same date. The paired rain panel exposes temporal bursts that monthly totals combine. The largest rain band mean, 95.20 mm on September 16 near -67.27 degrees, belongs to a particular averaging column, not

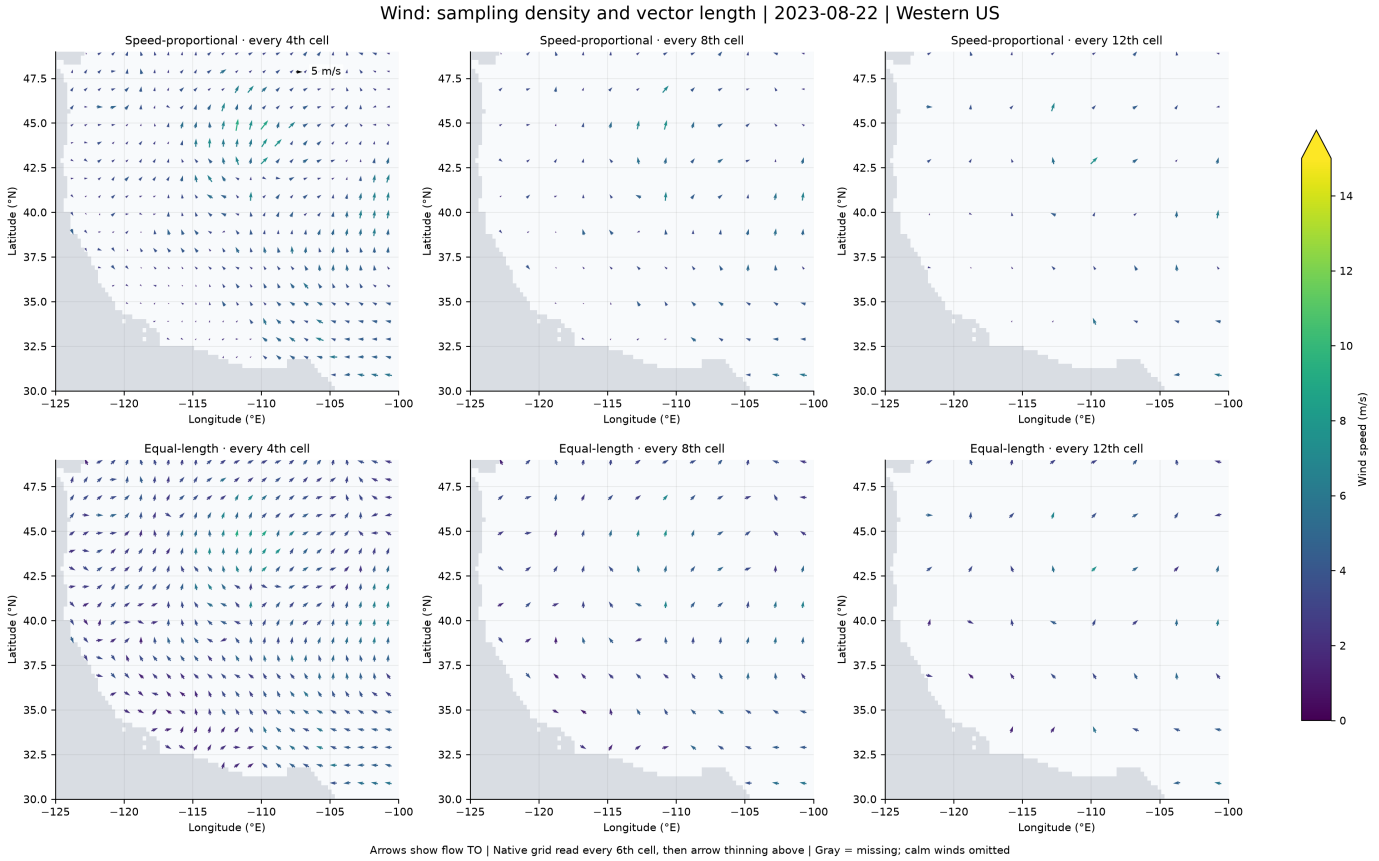


Fig. 25. Wind density and arrow length.

to an identified storm or the full country. Columns near the footprint edge may contain fewer cells; their support counts are saved for inspection. Latitude averaging deliberately sacrifices local detail to reveal time-space organization.

Limits. Longitude columns contain different amounts of valid land, and the support counts are saved with the arrays. Averaging across latitude hides local extremes. Apparent diagonal patterns cannot be equated with tracked storm motion. This is a descriptive surface-weather adaptation of a Hovmoller diagram, not an upper-atmosphere wave diagnosis or a climate-trend result.

Takeaway. The paired diagrams preserve when and where daily temperature and rainfall patterns occur, which monthly maps suppress. Their values are averages over a fixed latitude band, so they cannot identify local extremes or track individual storms.

IX. Wind: Retain Spatial Direction and Compare Monthly Distributions

The quiver experiment compares spatial sampling and length encoding on the same western-US wind field. The monthly roses summarize all valid sampled cells in the common full-domain footprint. These views have different spatial support and complementary purposes; their legends state those scopes.

A. Wind density and arrow length

Question (Fig. 25). Which arrow density and length encoding best reveal daily wind direction and magnitude?

How to read the graph. Columns retain every fourth, eighth or twelfth display-grid cell; rows compare speed-proportional length with unit-length direction arrows. All panels share the same western extent and speed-color limits. Arrows show flow-to direction after conversion from the source's meteorological from-direction. Colors still show speed when lengths are equal.

Detailed interpretation. The three columns thin an already subsampled field, so 1-, 2- and 3-degree arrow spacing should not be confused with native weather resolution. Sparse arrows reduce overlap but can remove local directional structure. Equal-length arrows recover the visibility of weak-wind directions while deliberately discarding length as a speed channel; color then carries that information. Because the same date, extent and speed scale are used, these differences can be attributed to drawing choices rather than different weather conditions.

Limits. Compare density within a row and length convention within a column. A large equal-length arrow need not indicate strong wind; its color carries that information. Longitude/latitude display corrections address direction distortion, but the output grid's fine spacing does not overcome gridMET's coarser wind input resolution.

Takeaway. Thinning trades directional detail for legibility, while equal lengths make weak-wind directions visible at the cost of a length-based speed cue. Read magnitude from the key and color convention, not arrow prominence alone.

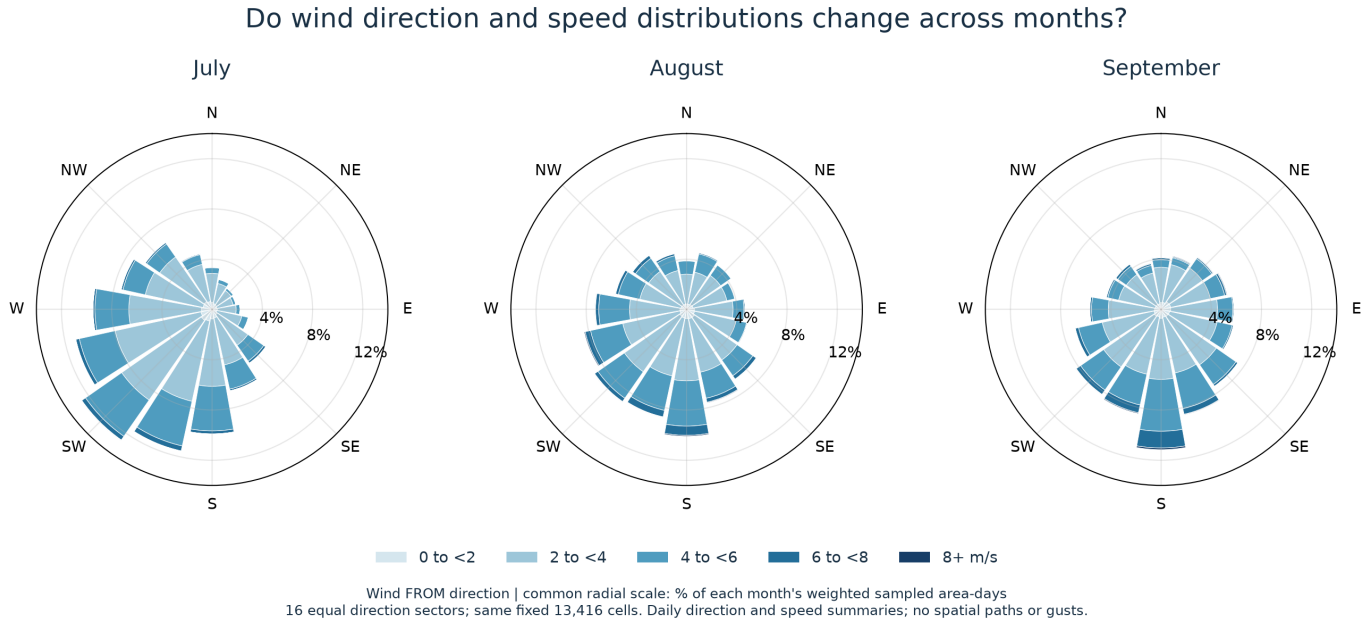


Fig. 26. Monthly wind roses.

B. *Monthly wind roses*

Question (Fig. 26). Does a pooled wind summary conceal monthly direction and speed differences?
How to read the graph. Each rose aggregates one month over the same full-domain 13,416-cell footprint. Sixteen equally spaced sectors show source FROM-direction, clockwise from north. Stacked colors classify speed into 0 to below 2, 2 to below 4, 4 to below 6, 6 to below 8 and at least 8 m/s. Sector radial height is percentage of the month's weighted area-days. All panels share a 14% radial limit.
Detailed interpretation. The largest directional frequency sector is southwest (225 degrees) in July and south (180 degrees) in August and September. The pattern therefore changes across months instead of being fully described by one pooled rose. The stacked bins show the associated speed composition. These modal sectors are not circular or vector means, and a domain-wide frequency cannot identify where each direction occurred.
Limits. Read radial height, not wedge area, as proportional frequency. Near-calm observations are retained in the lightest bin; their direction has little physical meaning. The domain and denominator differ from the western-box quiver. Daily summaries cannot resolve hourly wind durations, gusts or air-parcel trajectories.

Takeaway. Shared-scale monthly roses reveal a shift in the most frequent source-direction sector while retaining speed composition.

X. Joint Conditions: Distinguish Overlap, Location and Persistence

Separate variable summaries cannot establish their joint frequency. Aligned maps provide spatial context, exact set intersections quantify daily overlap, and run lengths recover the ordering of qualifying days. Thresholds are descriptive definitions, not official hazard classes, and daily aggregates do not resolve hourly simultaneity.

A. *Temperature, rainfall and wind in one aligned weather comparison*

Question (Fig. 27). Where do temperature, rainfall and wind patterns coincide on objectively selected contrasting days?
How to read the graph. Background color is daily maximum temperature. Blue labeled lines are precipitation contours at 5, 20 and 50 millimeters. Arrow bearing gives flow-to direction and arrow length gives speed, with a 5 m/s key. Dates define panels. The four measured fields therefore use separate channels; longitude and latitude provide spatial context. Every ninth cell of the already stride-six cache is retained for arrows.
Detailed interpretation. July 27 and September 30 were selected by a reproducible criterion: the highest and lowest sampled-area fractions meeting the hot/dry definition. Their 34.56% versus 1.75% values quantify why the dates contrast; the maps then show where temperature, rain contours and vector fields coexist. This does not establish that wind produced the difference. Tmax is a daily maximum, rain is a daily accumulation, and wind is another daily product, so temporal alignment by date must not be confused with simultaneous observations of a weather event.
Limits. The selection deliberately contrasts extremes and is not representative of every day. Daily maximum temperature, rainfall accumulation and wind summaries need not occur simultaneously. Arrow bearings account for map aspect and cosine latitude; they indicate where air flows, opposite the stored source direction. Coarse wind inputs and display thinning limit local inference, and overlapping patterns do not establish causation.

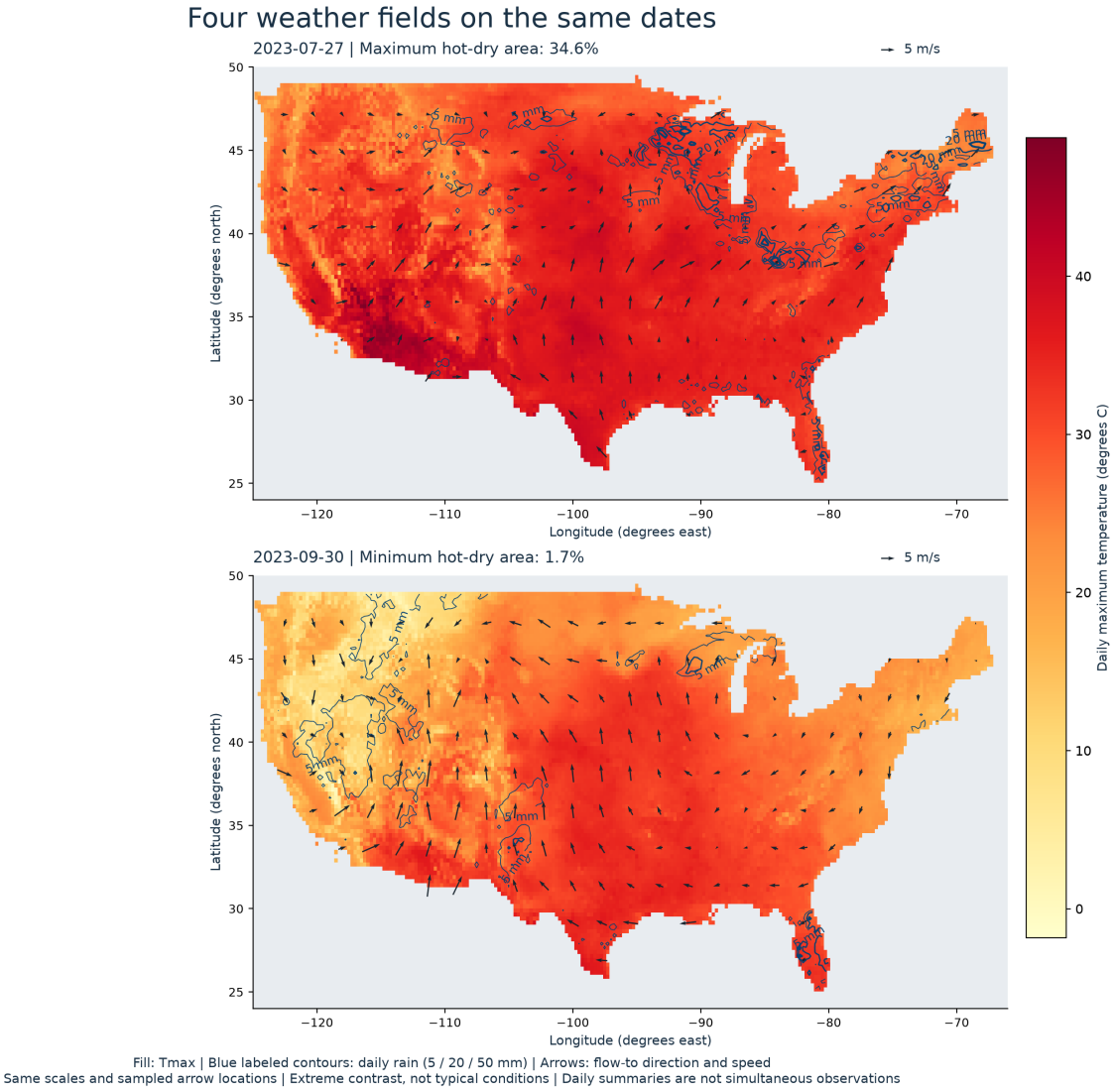


Fig. 27. Temperature, rainfall and wind in one aligned weather comparison.

Takeaway. The objectively selected dates contrast 34.56% versus 1.75% hot/dry area, while four separate channels expose their spatial context. Co-located daily temperature, rain and wind summaries support comparison, not event attribution or causal explanation.

B. Weather condition intersections

Question (Fig. 28). How often do hot, dry and windy daily conditions occur separately and in combination?

How to read the graph. Each column gives one exact intersection of three sets: Tmax at least 35 C, rain below 1 mm, and wind speed at least 5 m/s. Filled dots mean present and pale dots absent. Top bars are cosine-latitude-weighted area-day percentages; left bars show marginal set percentages. All eight combinations, including none, sum to 100% over the fixed 13,416-cell, 92-day support.

Detailed interpretation. Hot, dry and windy sets occupy 18.06%, 76.65% and 11.65% of weighted sampled area-days. Dry with neither other condition occupies 54.11%. Hot and dry with any wind occupies 16.27%, whereas all three conditions occupy 2.57%. Thus the large dry marginal does not imply the three conditions commonly occur together. Exported alternatives cover 33/35/37 C, rain cutoffs 0.5/1 mm and wind thresholds 4/5/6 m/s; overlap changes with the definition.

Limits. These are descriptive daily conditions, not official hazard classes. Daily Tmax, precipitation totals and wind summaries do not prove simultaneous hourly conditions. The chart pools time and space, so it cannot locate the overlap or distinguish isolated days from long runs. Neighboring cells and dates are dependent; no significance or independence claim is made.

Takeaway. Frequent individual conditions can have much less frequent joint overlap: adding the chosen wind threshold reduces hot/dry overlap from 16.27% to 2.57% of area-days.

C. Joint hot-dry frequency, persistence and threshold sensitivity

Question (Fig. 29). Are repeated hot-dry days scattered occurrences or sustained runs at the same locations?

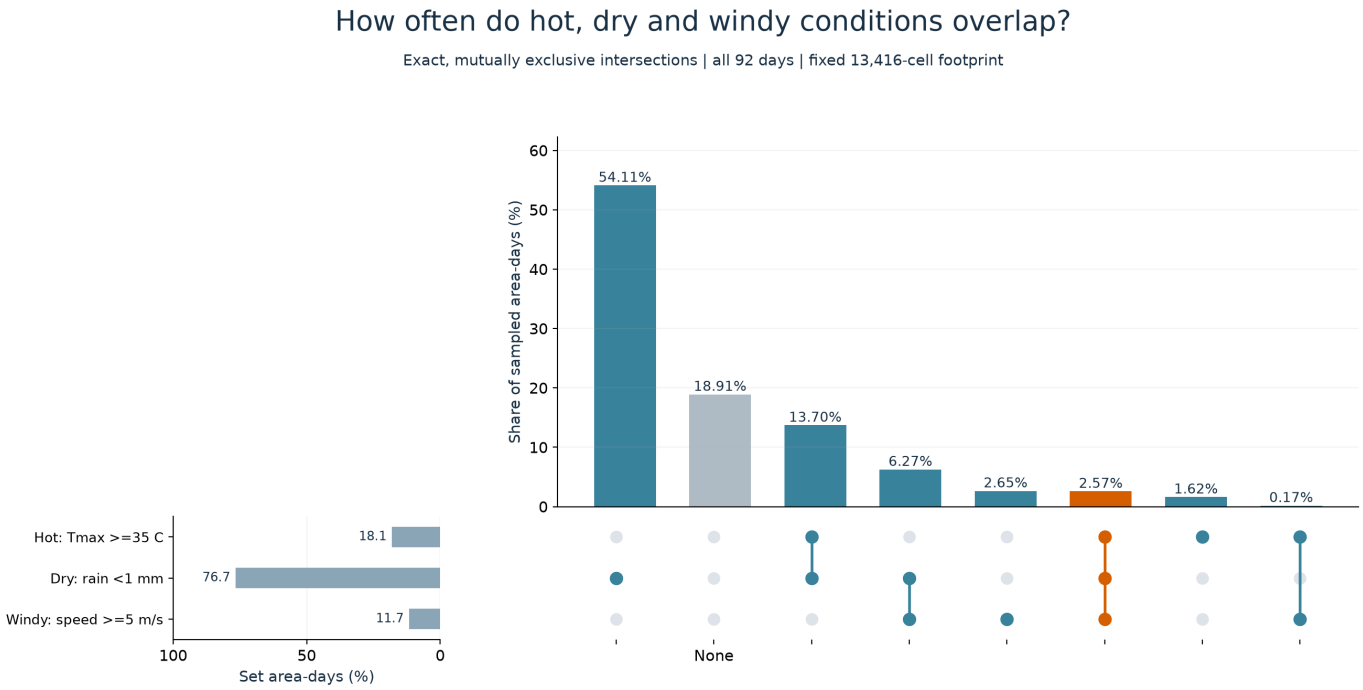


Fig. 28. Weather condition intersections. Method reference: [13].

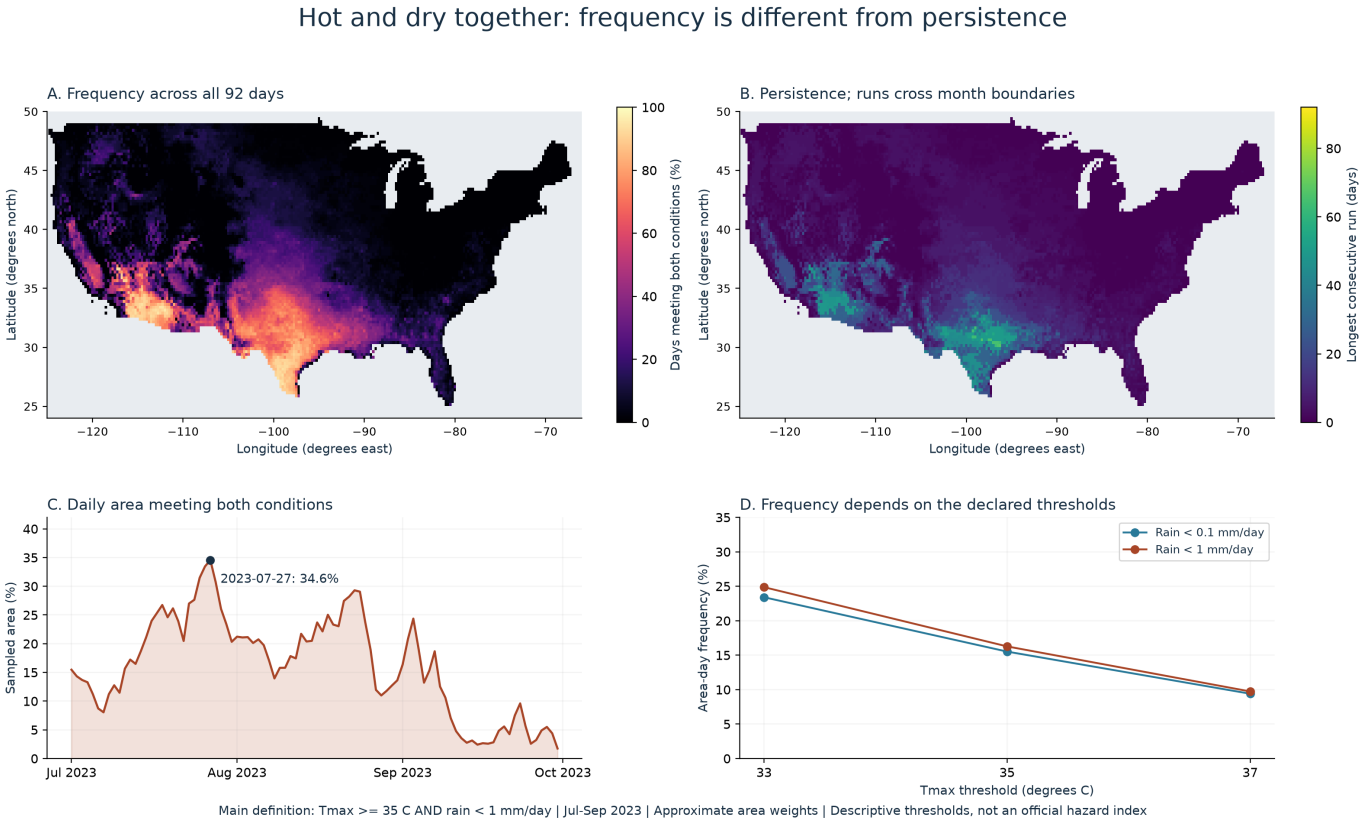


Fig. 29. Joint hot-dry frequency, persistence and threshold sensitivity.

How to read the graph. The first map shows the percentage of the 92 days satisfying the joint condition at each cell. The

second shows the longest consecutive run using the same daily Boolean sequence. The timeline gives each day's cosine-latitude-weighted area fraction; the final panel recomputes total frequency under six threshold combinations. Missing cells are gray; zero frequency remains a valid observed result.

Detailed interpretation. A frequency percentage counts qualifying days but ignores their order. The run-length map restores that information and finds a maximum continuous run of 67 days within the observation window. That maximum applies to at least one sampled cell, not the entire footprint; the mean joint area-day fraction is only 16.27%. The sensitivity panel shows how frequency decreases under stricter temperature thresholds and increases under a more permissive rain cutoff. Thus the detailed numerical result depends on the declared definition even though the distinction between frequency and persistence does not.

Limits. A maximum run can be truncated by the beginning or end of the observation window. The thresholds are descriptive and do not define official drought, heatwaves, fire danger, or population exposure. Spatial grid values and neighboring days are dependent. Approximate area weights describe the sampled land footprint, and no long-term climatic baseline is available.

Takeaway. Hot/dry conditions occupy 16.27% of sampled area-days, yet a location can sustain a 67-day run. Frequency alone hides concentration in time; retain threshold sensitivity and distinguish a local maximum from a domain-wide experience.

XI. Discussion: what the evidence answers

Answer to the main question. Averages conceal who differs from the typical country or location, snapshots conceal temporal order, and separate variable summaries conceal joint occurrence. The selected graphs recover these distinctions while preserving the original measurements. Their value comes from the specific question each answers and the checks that keep the comparison valid.

Connectivity. In the exploratory network partition, groups with mean adoption near 86% have different fixed and mobile infrastructure profiles. This does not identify a causal route to adoption because the groups were constructed from those same indicators and change with graph settings. In the 175-country matched panel, 126 countries (72%) increase adoption while landline intensity falls. The pooled broadband/adoption level association (Spearman rho=0.831) contrasts with the matched-change association (rho=0.054), so the cross-country level relationship cannot simply be read as an explanation of temporal change.

Progress and inequality. The new quantile comparison shows the P90-P10 absolute adoption gap widening from 28.63 to 69.84 percentage points between 2000 and 2020. Progress therefore does not imply absolute convergence. On the fixed 159-country panel, 69 countries reach at least 75% adoption by 2020, while 19 remain below 25%. In the classified regional panel, Europe's 2020 median is 86.86% and Africa's is 31.89%. Progress is broad but uneven; neither upward category transitions nor a global median establishes that regional inequality has disappeared. Country-weighted comparisons describe included countries, not the distribution of the world's people.

Weather. The September-minus-July change in mean daily Tmax is -4.64 C over the weighted sampled footprint, with 99.66% cooler. An endpoint-only difference disagrees in sign over 18.26% of area. Hot/dry conditions occupy 16.27% of area-days, but a local run can reach 67 consecutive days. Adding the stated wind threshold narrows joint overlap to 2.57% of area-days. Monthly differences, overlap and persistence are distinct quantities with different denominators and cannot substitute for one another.

Representation and robustness. Network positions, matrix ordering, bundling, hierarchy, color limits, contour intervals, vector density and KDE bandwidth change what is easy to see. They do not create new observations. Fixed cohorts/scales, exact edge lookup, empirical summaries, threshold experiments and graph sensitivity help distinguish representation effects from measured variation. Interquartile segments and spatial quantile ribbons show heterogeneity, not confidence intervals.

Limits. The country indicators are observational national aggregates; omitted factors and direction of effect are unresolved. Complete-case panels exclude countries. The similarity graph depends on features, standardization and k. Weather sampling omits finer spatial detail; neighboring cells and days are dependent. Daily aggregates do not prove hourly co-occurrence, and gridded daily wind is not a gust or station record. Boundary runs may begin before or end after the observed window. A single three-month interval does not establish climate change, anomalies relative to a long-term baseline, or hazard attribution.

XII. Reproducibility, artifacts and contribution status

Rebuild. From the workspace root, run `python -X utf8 ass2/scripts/preview_lecture_extensions.py`, `python -X utf8 ass2/scripts/verify_quality_figures.py`, then `python -X utf8 ass2/scripts/build_story_report.py`. The last command composes the selected report, updates the figure manifest and reading guides, compiles pdfLaTeX twice and produces the source bundle. The original A1 country inputs and existing daily caches are needed for full analysis reproduction; source paths and cache manifests remain in the workspace README.

Artifacts. `output/pdf/DAS732_A2_Report_IEEE.pdf` is the current report. `report_latex/DAS732_A2_Report.tex` and `output/DAS732_A2_LaTeX_Source.zip` contain its editable source and all selected numbered PNGs. `output/quality_figures/index.html` provides the same selected reading order with full-resolution original images. Three GIFs under `images/scivis/` contain eight frames each. `output/lecture_previews/` retains graph/threshold sensitivity tables, memberships and figure exports. The selected report has 29 figures, with supporting checks retained in the companion

gallery; previous 32-figure material is preserved in `archive/before_story_revision/`.

Validation. The quality checks independently extract run lengths over all 13,416 cells, recompute monthly means and compare Hovmoller values. New preview checks verify fixed-panel transition margins, symmetric adjacency counts, exact weather-intersection totals, threshold sensitivity and per-month wind-rose normalization. Report checks verify selected figure order, linked files, PDF references, text/layout and GIF frame counts. JavaScript pages are served locally and pass syntax/asset checks; live interaction testing remains pending as described above.

Current authorship status. The user instructed that the current implementation be treated as a single-contributor working version. A later split among the three team members has not been established. The team names from A1 identify the project team; they do not assign completed A2 task ownership. Before submission, the team must record the actual person responsible for each distinct SciVis and Infovis task and any later changes. Shared preprocessing and planned speaker roles do not establish task ownership.

AI assistance disclosure for this working revision. Codex/ChatGPT assisted with lecture review, visualization proposals, exploratory analysis and graph generation, report structure and explanatory text, code edits and verification. This assistance extends beyond proofreading. Team members must review the code and interpretations and supply truthful individual disclosures consistent with their actual use; this paragraph does not certify review or personal authorship on their behalf.

XIII. Conclusion

Averages hide profile differences and unequal country experiences; snapshots hide temporal change and persistence; separate variables hide joint conditions. Matched cohorts, shared scales, exact intersections and complete daily fields make these distinctions visible. The accompanying JavaScript pages provide required interactive methods, and GIFs retain the required scalar animations. The current working implementation is treated as single-contributor work; the actual later team split and individual disclosures remain to be recorded.

XIV. Companion material

The companion gallery retains the adjacency matrix, worked horizon construction, KDE sensitivity and adoption alluvial. Their full explanations remain available alongside this report; they are outside the PDF page budget.

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